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Workpackage C (WPC)

Implementation – Enterprise Characteristics

Deliverable C2: Reference Methodological Framework for processing online based enterprise characteristics (OBEC) data for Official Statistics V.1

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Introduction

This document presents development work on a Reference Methodological Framework (RMF) for processing online-based enterprise characteristics (OBEC) data for official statistics carried out under the ESSnet Big Data II Workpackage C "Implementation - Enterprise Characteristics" (WPC).

The RMF's purpose is to describe, at a conceptual level, a complete OBEC statistics processing pipeline across the four main big data life cycle phases: data acquisition, processing, modelling and interpretation, and producing statistical indicators (dissemination). This framework is intended as a reference guide and template for national statistical institutes (NSIs) within the European Statistical System (ESS) when setting up their own OBEC processing pipelines. The RMF should be relevant whether NSIs are implementing through local systems, integrating shared components developed by other statistical agencies, or using shared processing system provided by ESS such as the Big Data Test Infrastructure.

The results achieved during the previous ESSnet Big Data I project serve as a foundation and inspiration for WPC team in the developing stage of the current RMF. The ESSnet Big Data I Webscraping enterprise characteristics workpackage achieved two main results: solutions for detecting websites maintained by businesses and solutions for detecting features of interest on those websites. The website detection (URL retrieval) solutions - including methodologies, processes and software implementations - are based on search engine queries and machine learning techniques. The features of interest on those websites include ecommerce activities, social media presence and job advertisements.
Experimental work on classification, including NACE code detection, was also completed.

WPC generalises and extends this work as the RMF for use in any ESS country interested in producing OBEC, while retaining sufficient flexibility to accommodate specific national contexts or use cases.

When producing OBEC, we suggest the development and communication of a national web scraping policy. ESSnet Big Data II participant countries should follow the ESS Web scraping policy template. The policy template can be adapted to the specific national needs.

This RMF document is organized as follows:

- Chapter I describes business context (statistical products, usecases, proof of concepts) of the OBEC data
- Chapter II details the big data processing life cycle on online-based enterprise characteristics
- Chapter III deals with application and information architectures for OBEC data
- Chapter IV describes functional production prototypes and implementation requirements at national and ESS level
- Chapter V explains OBEC in the context of traditional ICT surveys and their integration with business register data to compliment and enrich the existing official statistics
- Finally, chapter VI details data quality and metadata issues relating with OBEC data
- Chapter VII consists all annexes to this RMF

I. Business context

This chapter describes, at a conceptual level, the direct statistical products from the four main big data life cycle phases of online-based enterprise characteristics (OBECs) for the production of official statistics. Additionally, a parallel has been drawn between the OBEC statistical products and the classical statistical products of the ICT usage survey in order to better understand the basic concepts and their similarities/differences.

Producing official statistics on business characteristics usually involves a classical statistical survey and/or administrative data. The former having the downsides of being burdensome on the respondent and incurring high costs for the national statistical institute (NSI) or other national authority (ONA). Administrative data might not include all necessary variables needed for each statistical product and available variables sometimes suffer from a considerable time-lag. Nevertheless, the statistical business register (SBR) is usually used as a frame for sampling, non-response analysis and calibration, and estimation. Web scraped data from enterprise websites has the potential to mitigate the shortcomings in both data sources. Scraping data from websites involves negligible burden on the company (the respondent) and using up-to-date scraped data can result in timely enterprise characteristics. However, the inclusion of web scraped data for a concrete and sometimes already existing statistical product is definitely not straightforward. For instance, it might not be possible to uniquely link a webpage to an enterprise since a single enterprise might use many webpages or a single webpage is may be used by many enterprises.

The following sections describe general concepts and definitions when using web scraped data as a supplementary data source for statistical products. The focus lies on the statistical products defined by the usecases in ESSnet Big Data II workpackage “Implementation - Enterprise Characteristics” (WPC).

1. Statistical products

This section describes the statistical products that are a direct result of the big data processing life cycle on online-based enterprise characteristics (OBECs).

The main elements that form the statistical products are: the statistical unit, the target population and the observation variables (at input level), the periodicity and the statistical indicators (at output level). The standardised main concepts are the additional elements ensuring that disseminated experimental statistics are harmonised and comparable among ESS countries.

Four use cases are defined in the ESSnet Big Data II WPC:

1. URLs (*Uniform Resource Locator*) Inventory;
2. variables in the ICT usage in enterprise survey;
3. data driven discovery of emergent enterprise classifications;
4. experimental language statistics.

The first and second use case are set within the context of the ICT survey and can be directly implemented in the regular statistical production whereas use cases 3 and 4 have more of an experimental character, not necessarily leading to new statistics.

Use case 1

The URLs Inventory represents a database containing URLs for each enterprise in the target population, where there can be one, many or zero URLs for a given enterprise. The URLs Inventory might already be available in an SBR, through another data source or from a specialised data provider. It can however also be built up from scratch by searching for enterprises via search engines like Bing or Google. The search results can be used for trying to answer the two questions:

a. Does an enterprise have a website?
b. Which URL is most likely to belong to that enterprise?

Commonly, a web search will return several results leading to different base URLs for one enterprise. A base URL, for example www.xyz.com or www.abc.com, might be included in several search results. The web scraper usually need not go deeper than one level in the website. Usually the information can be found in a contact section of the webpage or on the main page.

From a given list of URLs web content can be scraped to determine if a certain URL belongs to a specific enterprise. If the retrieved information is consistent with the stored information in the SBR then the URL is valid; otherwise, the URL is replaced by the URL with the highest confidence score or removed if the score is below a threshold.

Since it is regulated that only relevant data should be stored for official statistics, the data generated in the process can be deleted once the URLs repository is built. If generic web scraping is not possible then extraction of URLs from another available dataset can be an option. In such a case subsequent methods and classifier development would become the focus.

Use case 2

Use case 2 focuses on the extraction of information from an enterprise URL equivalent to variables from the ICT usage in enterprises survey. Whereas the URLs Inventory can be built for all enterprises in the population the ICT usage in enterprises survey contains only enterprises with 10 or more employees in the given categories of NACE Rev. 2. The extracted online based enterprise characteristics (OBECs) should answer the following questions from the ICT usage in enterprises survey; the corresponding ICT survey question in shown in square brackets (e.g. [C9b]).

1. Does the website have any of the following: online ordering, reservations or bookings (e.g. does the website have a shopping cart)? [C9b]
2. Does the enterprise use social media? [C11]
3. Does the enterprise use Twitter for a specific purpose [C12]:

    a) developing the enterprise's image or market products (e.g. advertising or launching products, etc.) [C12a]
    b) recruitment [C12e]

c) others – any tweet that does not fit into the other two categories. [C12b-d;f]

4. Does the enterprise have specific features on the website like:

a) description of goods or services, price lists; [C9a]
b) possibility for visitors to customise or design online goods or services; [C9c]
c) tracking or status of orders placed; [C9d]
d) personalised content in the website for regular/recurrent visitors; [C9e]
e) links or references to the enterprise's social media profiles; [C9f]
f) advertisement of open job positions or online job application.

5. Is the enterprise working on upcoming/new phenomena, specifically related to artificial intelligence (AI) and machine learning (ML).

The last question is not included in the ICT survey questionnaire, but is designed as an additional characteristic in order to identify innovative enterprises that are using AI and ML in their current activity.

Use cases 3 & 4

Both usecases 3 and 4 deal with classification of enterprise URLs based on the usage of language and text displayed in the URLs. One of the main methodologies applied here is the use of natural language processing techniques like Word2Vec or Doc2Vec to represent scraped text from URLs as high dimensional vectors. Finally, clustering methods or models are applied to the vector representation to derive possibly new classifications regarding business activity of enterprises.

A detailed description of the use cases can be found in the annexes of this document.

a. Main concepts

The main focus of using web scraped data as a supplementary data source for statistical production lies in the process of extracting online based enterprise characteristics (OBEC). As the name implies OBEC can be seen as any attributes, linked to enterprises, that have been extracted using data scraped from webpages. For example, the enterprise's URL if it exists.

The scraped data might be directly available through another data provider or need to be collected by the NSI itself. The collection of scraped data requires some sort of scraping software which directly extracts the content of an URL or renders a webpage via a browser and extracts the rendered content. Depending on the OBEC different parts of a webpage might be relevant so directly scraping the content of interest, if it is identifiable, can prevent accumulating unnecessarily large datasets. Having the scraped data at hand, the extraction of OBEC involves either using a deterministic rule set, or applying statistical or machine learning. Both deterministic and learning approaches need to be thoroughly tested and should have reached a state of maturity, which enables the production of high quality statistics.

To summarize, the procedure for extracting OBEC can be split into two main parts:

1. **Scraping:** Retrieve information on enterprises by scraping the world wide web for instance the enterprise URL (Uniform Resource Locator)

2. **Extraction:** Extract OBEC using the scraped data from step 1.

The extracted OBEC, if successfully retrieved, can then be linked to existing units (enterprises) in the SBR and used in the statistical production process either directly or as a supplementary data source.

b. Unit type

There are several types of statistical units. The SBR is the only source for deriving statistical units. As a result, coherent economic statistics with consistency across sectors, countries and geographic areas, and over time can be achieved. Among the SBR units mainly information for *enterprises* and *enterprise groups* can be derived from the web.

Enterprise[1]: The smallest combination of legal units that is an organisational unit producing goods or services, which benefits from a certain degree of autonomy in decision-making, especially for the allocation of its current resources[2]. Thus, it consists of one or more legal units, or may contain a part of a legal unit.
In the last decade the legal complexity of the organisation of enterprises has increased. Enterprises that used to consist of one legal unit have split their activities across a number of legal units. These splits are made for issues to do with tax, liability, tariffs, etc. and are not solely related to the economic purpose of an enterprise.
In the identification phase of the unit the tax code/VAT number is mainly used to confirm the exact correspondence between websites and enterprises. Attention should be paid to cases where there is more than one legal unit within the same enterprise, since each of these will use its own tax code/VAT number.

Enterprise group[3]: A set of legal units bound together by legal and/or financial links under the same control, i.e. control that has the ability to determine general corporate policy[4].
All enterprises in a group are likely to have their own website, but it is more likely that there will be a website for the whole group where all the necessary information can be found more easily (probably together with links to any websites of the companies in the group). Furthermore, the enterprise group has to produce consolidated accounts and annual reports for all the enterprises in the group, thus in its website more interesting documents in PDF format could be found, like the consolidated financial statements where the assets, liabilities, equity,

[1] EU 1993 Statistical Units Regulation (Regulation 696/1993 of 15 March 1993)
[2] A unit is deemed to be an enterprise if it: a) operates the necessary factors of production (e.g. human resources, capital, technology, land and in particular management); b) accesses the necessary controlling system, e.g. an integrated cost calculation, which covers the main, secondary and ancillary activities of the unit deemed as enterprises, and c) has adequate managerial structures, i.e. managers that can decide about the production process and about the economic transactions.
[3] EU 1993 Statistical Units Regulation (Regulation 696/1993 of 15 March 1993)
[4] The control can be exercised by (a) owning more than half of the voting shares, (b) having the right to appoint or remove a majority of the members of the management, (c) having the right to exercise a dominant influence over it, (d) controlling more than half of the shareholders' voting power of another legal unit directly or indirectly, or otherwise (e) proving that there is de facto control exercised. Indirect control refers to controlling a legal unit via another legal unit.

income, expenses and cash flows of the parent company and its subsidiaries are presented as those of a single economic entity.

Although the prime use for OBEC is to substitute the survey response of an enterprise the unit an OBEC refers to may not always be the enterprise itself. One can consider the website of an enterprise as a response unit in order to derive a statistical estimate which refers to the URLs of enterprises in a certain target population. Considering the ICT usage in enterprises survey some observation variables refer not only to the website of the enterprise itself but also the website of the parent or holding company. Thus, a single URL can lead to OBECs for multiple enterprises. To put it in a nutshell the units for retrieving OBECs can be comprised of enterprises and/or webpages.

For comparison purposes, the enterprise as the unit for retrieving OBECs "is the smallest combination of legal units that is an organisational unit producing goods or services, which benefits from a certain degree of autonomy in decision-making, especially for the allocation of its current resources. An enterprise carries out one or more activities at one or more locations. An enterprise may be a sole legal unit" as defined in Council Regulation 7 (EEC) No 696/93 of 15 March 1993 on the statistical units for the observation and analysis of the production system in the Community. In the current RMF the terms enterprise/enterprises and business/businesses are used interchangeably.

The website of an enterprise as the unit for retrieving OBECs is defined as some digital solution (in the World Wide Web) that an enterprise has or offers to customers.

c. Population

The target population for official business statistics is usually all of the units in the SBR or a subset thereof, as is the case for the ICT usage in enterprises survey. In particular, when estimating statistical indicators from OBECs, one could derive an estimate based on the subset of the population formed by enterprises that have one or more URLs associated.

Target population

The first task is to correctly report internet data to population units, i.e. data collected from websites must be referred to the population unit "enterprise". The availability of the SBR, a population frame containing all enterprises included in the target population, suggests using it as a basis for a search of the corresponding websites.

It is often useful to focus on a specific area of interest. Depending on the purpose of the research, it may be convenient to limit the production of the OBEC statistics to a part of the population that represents a specific sub-population of the business universe, instead of relating to the entire SBR.

Depending on the statistical indicator in question OBECs might be retrieved for the whole target population or just a sample thereof.

Sampling population

Once the target population is assigned a sampling frame is then drawn from it. It has been observed that the difference in the structure and content of the websites depends to a large extent on the size of the enterprises, the complexity of their organization, and the economic activities they carry out. Therefore, in order to correctly evaluate the results of the web scraping activity it is recommended to extract a stratified sample from the SBR. This could take into account the disparities caused by these factors.

Because of their relevance, usually active enterprises in the reference year are chosen, having the legal form of a corporation or a partnership. To improve precision, a stratified sample of these SBR units could be extracted from the reference universe in several possible ways, for instance by using proportional allocation. This way the sampling fraction in each of the strata is proportional to that of the total population. It is suggested to consider a sample of enterprises stratified both by size, in terms of employment and turnover, and by economic activity in order to take into account the differences in the structure and contents of the web sites possibly caused by these factors.

The sampling population on OBECs comprises of enterprises with potentially related URLs having a scoring rate above a given threshold (scoring given by online search engine for each enterprise). This might also depend on the capability of the underlying production process, e.g. if it is technically feasible to scrape a very large number of webpages during the observation period. The statistical indicator could also be estimated using the OBEC from a sampled subset of the target population and calibrating sampling weights accordingly.

A register based approach to big bata means definitively that all statistical, administrative and web information will be 'catalogued' in the SBR for multiple purposes, thus assuring consistency and giving concrete support to statistical production based on the SBR. It will ensure proceeding in a structured and integrated manner, placing the register at the centre.

In this sense there is a "two-way" information flow, to and from the register. The two parts evidently benefit from each other's presence. On the one hand, the register acquires new information earlier from the web and catalogues it in a coherent way for multiple purposes, augmenting its content and ability to support statistical production. On the other hand, the data from the web, once inside the informational world of the register, is integrated with all the other variables from administrative and statistical sources. In this way the unstructured data from the web acquire a "structure" and could benefit from all the available statistical classification. The link with the register provides information on the subpopulations of interest and therefore on which is the reference universe of the data; they receive a name, a size dimension, a location in the territory, etc. using all the classifications in the register. The wide coverage of the SBR becomes a support platform for big data.

d. Periodicity

When using OBECs as an alternative data source for official statistics the time frame for retrieving the OBEC should be in accordance with the observation period as defined by the guidelines of the ICT usage in enterprise survey. However, the periodicity can be increased if scraping does not place much burden on the URLs visited. For certain OBECs, for instance the

URL Inventory, it would also be necessary to scrape and retrieve the URLs of a business more frequently in order to keep the inventory up to date for the retrieval of other OBECs.

Some issues like the updating and maintaining strategies are questions that still need a proper answer, like the matter of when and how it would be necessary to repeat the extraction of the same data from the web. Although very expensive, maybe a cyclic monitoring to capture differences could be useful. Regardless of approach, it must be understood when differences can be considered significant as true changes, and moreover how to get the date in which the change takes place, crucial for the SBR metadata.

e. Observation variables

Contrary to a classical ICT usage in enterprises survey, variables are not observed by collecting answers to a questionnaire but by using search engines, APIs and/or web scraping software to scrape data which potentially holds information on the variable of interest. From the scraped data OBECs can be extracted using either machine learning algorithms or deterministic rules and from one or more OBEC observations variables can be derived. For instance the variable *"does the enterprise have a website"* is a binary, 0/1 or "yes"/"no", variable which is derived by using the URLs Inventory which holds the URLs for each enterprise, which for a given enterprise could be one, many or zero URLs.

Other potential OBEC derived target variables (to be extracted from raw textual data scraped from webpages) could be the following.

- What do enterprises sell: main products/services sold.
- How do enterprises sell: selling channels (e.g. online shopping card, booking services, delivery services).
- To whom do enterprises sell e.g. business-to-business, business-to-consumer.
- Where do they sell: national/multinational markets.

f. Statistical indicators

The following statistical indicators were generated (defined in use cases 1 and 2):

- Rate of enterprises having websites
- Rate of enterprises engaged in web sales on their website
- Rate of enterprises that are present on social media
- Rate of enterprises using Twitter for a specific purpose
- Rate of enterprises having specific features of the website
- Rate of enterprises working on upcoming/new phenomena, specifically AI and ML

All statistical indicators listed above can be directly estimated from the respective OBECs.

In the following it is assumed that the OBECs were retrieved for a sample $e_1, \dots, e_n$ of the whole target population and that each enterprise $e_i, i = 1, \dots, n$ has a corresponding sampling

weight w_i.
The sole indicator for usecase 1 is the rate of enterprises having a website:

$$R_{web} = \frac{\sum_{i=1}^{n} w_i \, I[u_i \neq \emptyset]}{\sum_{i=i}^{n} w_i} \tag{1}$$

where $\sum_{i=i}^{n} w_i = N$ is the number of units in the target population, $I[.]$ is the indicator function and u_i is the set of all websites for unit i, which is empty if no website was found. This indicator can easily be subdivided by NUTS regions, NACE categories or a grouping by persons employed. Let $A_1, \dots, A_p$ be a partition of the whole target population $\mathcal{A}$ i.e.

$$\bigcup_{i=1}^{p} A_i = \mathcal{A} \text{ and } A_i \cap A_j = \emptyset \quad \forall i \neq j \tag{2}$$

then the rate of enterprises having a website for a specific group A_k is defined by

$$R_{web,k} = \frac{\sum_{i=1}^{n} w_i \, I[u_i \neq \emptyset \wedge e_i \in A_k]}{\sum_{i=1}^{n} w_i \, I[e_i \in A_k]} \quad . \tag{3}$$

Use case 2

In similar fashion the indicators for use case 2 can be defined, however the target population changes from all enterprise to enterprises having a website. Most of the indicators in usecase 2 can be defined by

$$R_{ICT,q} = \frac{\sum_{i=1}^{n} w_i \, I[u_i \neq \emptyset \wedge V_i^q = 1]}{\sum_{i=1}^{n} w_i \, I[u_i \neq \emptyset]} \quad . \tag{4}$$

with V_i^q equal to 1 where the OBECs for enterprise e_i fulfills question q. q can be any of the following questions

- Does the website have any of the following: Online ordering, reservation or booking (e.g. shopping cart)?
- Does the website have links or references to the enterprise's social media profiles?
- Does the enterprise have specific features on the website?
- Does the enterprise work on upcoming/new phenomena, for example: AI and ML?

For the indicator on using Twitter for a specific purpose the target population needs to be adapted to hold only enterprises for which the OBECs on the usage of social media is equal to 1.

It could happen that for certain enterprises $e_{i(1)}, \dots, e_{i(k)}$ not enough data was scraped such that the corresponding OBECs do not fulfill one of the above questions. To eliminate technical root causes it is recommended to use a scraper which can deal with embedded JavaScript, for example Selenium, and to try to reach webpages at multiple different times. Even considering technical issues the scraper might still not scrape the necessary information since the structure of webpages is not standardized, leaving the procedure to silently fail extracting OBECs. For the unlikely event that it is known that the scraping was not succesful for a subset $e_{i(1)}, \dots, e_{i(k)}$

of the sample, these cases can be considered as non-response and treated accordingly, for instance by adjusting the weights.

2. Current statistical production processes affected

In recent decades the combined use of data from different sources for statistical purposes has become a consolidated practice. Along with the use of administrative data for setting up statistical registers, the experimentation of the use of big data has been ongoing for some time in order to update, add, and validate information in the SBR. The most used source for finding big data is the web. The vast amount of information available presents new opportunities, but also new challenges for data integration experts, given the structural differences between administrative and internet data.

In administrative sources the identification of the unit is sure. In addition, the integration of the unit's data is simple, as they use a common identification code (tax code, VAT number). Through a consolidated process of integration, an ID code for legal units and possibly an ID code for enterprises are assigned.

In contrast to administrative data big data presents serious risks. Some of them are evident, like the difficulty to manage rapidly growing volumes of data resulting in a high consumption of computing and storage resources. Moreover there are technical limitations to solve, like the long run time necessary for the crawler to get all the content and the restrictions caused by the security barriers of the website preventing automatic access. Others are statistical problems, such as: the difficulty of certifying the quality of information and the data reliability, how to attribute information with certainty to an SBR enterprise, and how to classify with the universe to which a set of data extracted from the web belongs with certainty.

Despite risks there are also obvious benefits such as enriching statistical production with new information, improving the timeliness of statistical products and increasing the relevance of business statistics at a lower cost compared to enlarging the existing data. A point of strength is that the web is an independent source of data, while all the other sources, administrative and statistical, can be considered linked in some way and influencing each other. The picture provided by the data may be a representation closer to reality: this is how the enterprise sees itself and how it wants to present itself to the outside.

It is a new idea to use big data as an additional source for the SBR, through web scraping and text mining technologies, with the aim of integrating the 'structured' business data with the 'unstructured' data coming from web pages.

Indeed there are many difficulties related to the use of these new sources. Big data techniques changes the way we collect, analyse and integrate data. The added value lies exactly in the information that is hidden in the data, and in the proactive use of the data, namely data-driven, that is, reading the data and using them as a starting point to create a strategy. Therefore to integrate the big data analytics tools in the context of current production processes is a difficult

task. This means to combine data-driven processes – based on input data that do not come from sources that are specific for statistical purposes, as they are not homogeneous, not structured or semi-structured, and not stable over time – with processes based on an output-oriented approach, since in the context of official statistics, production processes are constructed with a view to obtaining statistical outputs.

The internet as data source represents new opportunities and challenges for official statistics, which should incorporate all innovative potential data sources as much as possible into their conceptual design. An increasing number of national statistical institutes are indeed experimenting with the use of new sources of data, also known as big data, in order to produce the same or new statistical information in a multisource environment more efficiently and with higher levels of quality. There are important tasks yet to be faced with the use of big data and in general for Statistical Institutes. The main challenge for using big data is to move from experimentation to production. This step involves various aspects ranging from respect of privacy to the need to acquire new infrastructures (methodological, technological, organisational), as well as new skills.

The web scraped data on OBECs from enterprise websites have the necessary potential to be an additional source for producing official statistics. This is also the main objective that is achieved through the execution of the first two use cases (described in the Section 1, Chapter I), namely: to make a decision to use the new information to produce even more detailed statistics on ICT use.

This section briefly describes the statistical business process, the basic concepts and definitions of the ICT usage in enterprise survey.

The main statistical production process affected is the regular ICT usage in enterprises survey where the web scraped data is a qualitatively new type of data source that can supplement missing data, provide more timely information, or even explore new phenomena in the digital economy. In addition, web scraped data could be used for verification and control purposes of survey data.

Eurostat is responsible for coordinating the ICT usage in enterprises survey that are conducted at national level according to the Regulation (EO) № 808/2004[5]. The survey is carried out annually according to a set of regulations and it is based on model questionnaires and the accompanying methodological guidelines for their implementation.

The observation unit in the ICT survey is the enterprise, as defined in Council Regulation 7 (EEC) No 696/93 of 15 March 1993 on the statistical units for the observation and analysis of the production system in the Community. The regulation defines an enterprise as a kind of statistical unit: “The enterprise is the smallest combination of legal units that is an organisational unit producing goods or services, which benefits from a certain degree of autonomy in decision-making, especially for the allocation of its current resources. An

[5] https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32004R0808

enterprise carries out one or more activities at one or more locations. An enterprise may be a sole legal unit."

The target population of the survey is the group of enterprises delimited by the following attributes: economic activity and enterprise size. It is mandatory to observe enterprises with 10 or more persons employed whereas is is optional to survey enterprises with 0-9 persons employed.

In the survey on ICT usage in enterprises, the observation variables are mainly (binary) categorical variables. Some characteristics to be collected through the ICT survey are the same as OBECs presented, for instance: having a website, provision of the following facility: description of goods or services, price lists, online ordering or reservation or booking, links or references to the enterprise's social media profiles and etc. It is important to mention here that the observation variable "having a website" refers to the use of a website by the enterprise to present its business, excluding the presence of the enterprise in directories, online yellow pages and e-marketplaces.

The web scraped data on OBECs also affects the updating of the SBR. The characteristics of each unit in the SBR can be supplemented by OBECs through matching the units by unique identification number or by combining several features. This enriches the frame population for all statistical business surveys.

The enterprise websites should be analysed by means of the big data analytic techniques in order to produce a set of new enterprise-level information linked to the SBR, whose content allows:

- to complete the missing information in certain variables of the Business Register;
- to check some information for variables enclosed in the Business Register;
- to add new information to cover additional variables for the Business Register.

The new information can be classified according to the type of business enterprise characteristics extracted from the web. In particular, the following types can be considered.

- **Identification and contact variables** – *main personal characteristics of enterprises*: they are related to the specific individual features of the enterprises, such as any graphical characteristics and personal data. They are mainly useful for the correct identification of the company on the web.
  - *URL-address (Uniform Resource Locator):* the address that shows where a particular page can be found on the web. It may already be available before starting web scraping or it must also be obtained directly from the web. The acquisition of the enterprise's web address could be:
    - *i.* from the administrative sources used to update the SBR (usually available only for a low percentage of SBR enterprises, it must be checked by a validation procedure);
    - *ii.* downloading information from some proper thematic directory sites (these URLs are the result of a specific web scraping process to extract the identification variables of enterprises from some thematic business portals);

iii. performing batch queries on the search engines by means of the available SBR enterprises identification characteristics (URL retrieval with machine learning techniques);

iv. obtained as a result of a process of extracting URLs from the certified email address contained in administrative sources (using a procedure that extracts the domain of the enterprise website from certified email and verifies if a correspondent valid URL exists).

- *Tax fiscal code and VAT registration number*[6]*:* they are defined nationally, often by the tax administration, and do not exist in all countries; certain economic activities may be exempted from VAT and in this case another administrative identity number can be used, e.g. general tax number or corporate tax number.
- *Business name:* legal name, family name if natural person, trading name, etc.
- *Business street address:* physical address, including postcode and official geographic code at the most detailed level.
- *Email address, certified email address:* this enriches the SBR with updated information that is useful for survey managers and supervisors for communication with the enterprises.
- *Telephone and fax number:* this information is very useful since it is not well covered and updated in the administrative sources. It is used in the check phases of all the business surveys and it is also useful for SBR staff when contact with an enterprise is required.

This first group of variables should be used for the *identification of the enterprises on the web*. Usually in the SBR the information regarding the identification characteristics of the enterprises is very accurate, as it can be easily updated quickly using administrative sources. Then the need to obtain their latest updated values from the web arises from the potential use of these variables for the correct identification of the company on the web, which is a first step to be carried out before using data from web scraping.

In whatever way the web address has been found it is considered crucial to proceed with its identification in order to confirm the exact correspondence between websites and enterprises in the SBR. When the URL is available it is useful to run first some procedure for its syntax validation, including a check and correction of recurring errors. In all cases, the detection of all the possible existing identification variables from the website through the use of information retrieval techniques should be performed for the recovery of the relevant textual information contained in the extracted text. In particular the retrieved information is compared with the same characters available in the SBR register through matching techniques and string similarity metrics.

Since the tax code/VAT number allows unique identification of the enterprise, it is important to highlight that in case of a difference between the values extracted from the website and those

[6] The main problem with the VAT number is that it may not relate only to one legal unit, but to a group of legal units which may equal an enterprise or an enterprise group or its subgroup. In such cases, the VAT number can be the same for several legal units. In some countries, it may also relate to part of a legal unit or to a natural person who is not an economic operator, but who may have a VAT number for tax reasons.

contained in the register it is necessary to investigate in more depth if a demographic event happened that could justify this occurrence.

- **Structural variables** – *main economic/stratification characteristics of enterprises:* they are related to the structural features of the enterprises such as the size or the main activity of enterprises. They are mainly used for sampling survey frames, grossing up sample data and/or presenting results.
  - *Legal form* (it is the legal status of the unit; the possible types differ across countries according to the particular legislation in each country; mainly they are sole proprietorship, partnership, limited liability company, cooperative society, non-profit making body, publicly-owned enterprises, etc.).

    As legal a form is required in all business tax obligations, this is already high quality information in the register and does not need to be updated by the web scraping process. For this reason it is often used in the same way as the identification variables described above: as an additional variable for the identification phase.
  - *Principal economic activity and secondary economic activities*: they are used in determining the probabilities of units being included in survey samples, it is therefore important to take note of this information as soon as it changes, even if the change is not applied in the SBR for its stability rules (the use of stability rules is recommended by the NACE regulation in order to suppress movements that are no more than temporary phenomena or statistical artefacts).

    This is especially applicable for large and complex units, whose changes in performed activity have more impact on statistics so there is an increased incentive to ensure the changes are correct. It is not recommended to directly use the value coming from the web, rather potential updates should be checked by comparing data from all available sources to see if there is consistency, or by contacting the units concerned directly.

    Potential updates to small units have less impact on the resulting statistics and for that reason, the procedures for their maintenance can be largely based on automatic updating from the web without any need of confirmation.

    There are several possible ways to obtain information on the economic activity performed by the enterprise using data from web scraping.

    Otherwise the general short description of the business activity that each search engine provides in the line of results could be used. This represents a descriptive showcase of the website.

    Another approach is the classification of enterprises into macro-categories of products using the list of products eventually included in the web page: information on the type of products used in the three phases (input, process and output) of the production process can somehow be treated to create macro-categories, useful to validate the main and secondary economic activity in the SBR.
  - *Turnover* (for production statistics, value added is considered the most appropriate characteristic but it is difficult to measure and is often replaced by proxy measures.

Because it is relatively easy to collect, the most commonly used monetary proxy is turnover (revenue or sales).

- *Company capital.* This information only refers to those companies that publish their share capital on their website. This type of information is also useful for creating new a dimensional class that measures the companies by risk capital.

- **Other qualitative variables** – mainly qualitative characteristics of enterprises:
  they concern information that isn't directly measurable but typically representative of some characteristic of the enterprise, such as:
  - *Links or references to the enterprise's social media profiles*. Identifying companies on social media (that use these new tools for marketing activities) provide a new channel for getting more information from enterprises. This information can be considered as an indicator of the use of IT for marketing.
  - *The presence of online job application facilities*. It identifies enterprises that publish job application facilities on their website as well as allowing the enterprise to find specialised profiles by using web technologies and provides another useful skill-based indicator for classifying the economic activity of companies.
  - *Links to other specific site pages.*
  - *The identification of documents in PDF format.* The two most relevant are those relating to balance sheets (financial statements) and product catalogues. Financial statement allows for the extraction of an enterprise's characteristics and other relevant information about economic activity. It's also useful to profile the enterprise while the product catalogues provide useful information for better classifying the economic activity performed.
    In general, all the documents in PDF format could be subject to clustering procedures with the aim of dividing them into proper thematic classes. This can be achieved by using complex text mining techniques based on the textual content of the documents.
    Otherwise, when the content – and accordingly the class – is already known (for example, in case of clear recognizable financial statements, consolidated financial statements, product catalogues, policy documents, etc.) it is useful to search for keywords or other characteristics of the classes for their cataloguing.
    In both strategies of categorising the PDF documents, they must be analysed with appropriate techniques to use their content in order to increase the information contained in the register.
    - *New business taxonomies.* Applying text mining and machine learning techniques to the content of the websites for a subset of enterprises for which the specific characteristics of interest are known (e.g. belonging to the top-level NACE category), it is possible to use this training set for the construction of patterns able to assign a score to this phenomenon for all other enterprises. It is a feature engineering approach able to identify elements that can catalogue the company in a new taxonomy for which there are no well-defined qualifying requirements to establish

whether or not an enterprise falls into this category. With these operations it is possible to read, measure and grasp aspects of the country's economic activity that are not yet observable. In particular this goal could be probably reached as an output of the use case 3 (Data driven discovery of emergent enterprise classifications) within which new pseudo-topics will be identified by testing different algorithms such as word2vec and t-SNE (t-Distributed Stochastic Neighbor Embedding) plus k-means.

A proper SBR maintenance strategy implies constant updating of the SBR, not only in terms of coverage but also concerning content. Changes in the values of the characteristics of existing statistical units have to be identified and recorded. To establish and maintain the values of the characteristics it is required to record useful metadata associated to each value, at least the source of the new value, the date of the change and of the update, the reference date and the previous value. These are valuable in helping assess the quality of the values of the characteristics and in auditing change processes.

II. Big data processing life cycle on online-based enterprise characteristics (OBEC)

This chapter describes enterprise characteristics web scraping processes in a generic way. A generalized process for detecting enterprise characteristics from web data is presented, drawing on experiences from this project so far and the previous ESSnet Big Data I. This generalised process provides a starting point for any ESS country that would like to derive online-based enterprise characteristics in their national context.

The first section reviews processes from the more general viewpoint of the big data processing lifecycle defined in the ESSnet Big Data I. The second section presents a high level, generalised sketch of enterprise characteristics web scraping processes and explains relevant GSBPM[7] 5.1 phases in more depth.

1. Enterprise characteristics big data processing lifecycle

The process of web scraping for detecting enterprise characteristics naturally fits into the more general process of the big data processing life cycle, as specified in the ESSnet Big Data I project:

> "*The big data processing life cycle focusses on the part of the Big Data process immediately following the generation and availability of Big Data. During the big data life cycle, the data is* ***checked***, ***transformed***, ***aggregated*** *etc. and –in the end-* ***turned into information****. This process is* ***not linear*** *and contains many* ***cycles*** *as findings at the*

[7] Generic Statistical Business Process Model, https://statswiki.unece.org/display/GSBPM/GSBPM+v5.1

*end of the chain may initiate the need to improve some or a number of previous steps. Big Data can be processed in a **batch** but also in a **streaming** way.*"[8]

Many of the generic characteristics of the big data life cycle sketched here do indeed apply to the process of scraping for enterprise characteristics:

- Retrieved data is ***checked*** (validated) against phase-specific criteria during both the URL retrieval phase and the web scraping phase;
- Raw scraped data is ***transformed*** into texts for analysis and interpretation in order to derive meaningful indicators on enterprises;
- As with any national statistical production, microdata is ***aggregated*** before publishing e.g. to NACE class, number of employees or NUTS region;
- In general, the whole enterprise characteristics web scraping process is intended to exploit the rich web to turn it into meaningful ***information***. This is not merely the bare statistical aggregates published in tables, but also leads to new knowledge on the internet economy as a whole, as was shown in an early study by Statistics Netherlands.[9]
- Indeed, the process is not ***linear*** but ***iterative***, encompassing ***cycles***. Matched URLs can be merged into the statistical business register, obviating the need for a URL retrieval phase in consecutive runs (although care should be taken to evaluate this regularly). Discovering enterprise URLs via search engines may require multiple search runs, comparing results with already known enterprise data at each run. Web data interpretation may lead to new evidence on the reliability, or lack thereof, of an enterprise/URL association, which this may lead to a new URL retrieval cycle run.
- A business register may contain millions of entries. Depending on the rate of enterprises for which URLs are already known and the number of search requests required per business, URL retrieval may involve long running ***batch*** processes to complete. The same holds for the scraping phase. Processing of ***streaming*** data is less relevant here but might apply if streaming data from social web (for example twitter) is considered during the enterprise characteristics determination process.

All in all, we conclude that many - or all - of the characteristics of the generic big data processing life cycle apply to the enterprise web scraping process. It is in all aspects a big data process and needs to be treated as such.

The next chapter explores relevant big data processing life cycle sub processes, which are expressed in terms of GSBPM phases: Collect, Process, Analyse and Disseminate.

[8] Deliverable 8.3 of WP8 Methodology of ESSnet BD entitled "Report describing the IT-infrastructure used and the accompanying processes developed and skills needed to study or produce Big Data based official statistics"
[9] Measuring the internet economy in The Netherlands: a big data analysis, Oostrom et al., Statistics Netherlands, 2016

2. Description of the big data processing life cycle: web scraping phases

Before diving into subprocesses, the enterprise characteristics web scraping process is sketched at a high level. To facilitate applicability in any ESS country, this is presented as generically as possible. We make the following assumptions:

1. Any national statistical office has some version of a *Statistical Business Register* (SBR). This is often a physical statistical register maintained by the national statistical bureau, but in some cases may be based on a register maintained by another governmental administration, such as the revenue/tax authorities, and used for statistical purposes. At a minimum, SBRs contain business names and one or more national identification numbers. Some businesses websites may already be known. To be generic we assume that the rate of enterprises with a known URL may vary between 0 and 100%.

2. Every country has some national or international search engine(s) that can be used in the URLs retrieval phase.

With these assumptions, we can sketch a general high-level view on the process of determining enterprise characteristics. Figure 1 shows this view:

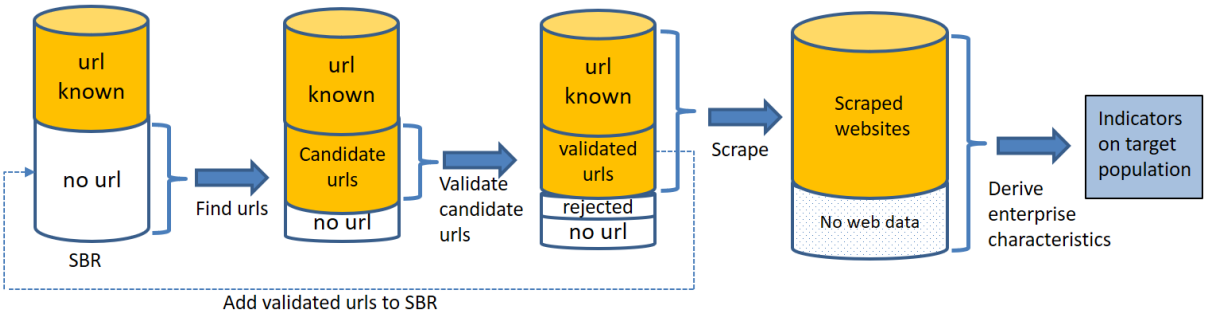


Figure 1: High level view on enterprise characteristics web scraping process

The phases are as follows:

1. URL retrieval for businesses without assigned URLs. Searches result in ***candidate URLs*** for this set. Note that it is common that there are still enterprises for which ***no URL*** can be found. They might not have any website, or it could be that our search strategy cannot find it.

2. ***Validation*** of candidate URLs based on comparing enterprise known SBR data with data from search results. Some candidate URLs will be ***rejected***. As stated in the previous section, the URL retrieval and validation/rejection can be iterative.

3. Scraping phase. Both the enterprises for which the URL is already known in the SBR as well as the validated URLs that have been found are used to scrape web content and store the ***scraped websites***.

4. ***Derive enterprise characteristics*** for the target population. This can be the whole SBR or a sub population, such as the ICT usage in enterprises survey population. Note that businesses for which there is no associated web data must also be accounted for in this estimation process, for instance by calibrating using known totals in SBR.
5. ***Updating the SBR with validated URLs to the SBR*** feeds into consecutive iterations and other statistical processes in addition to the main loop. This is shown as an arrow from the validated URLs to the SBR. Note that it may be worthwhile including an indicator as to whether a URL originates from an administrative source or from the URL retrieval process in the SBR.

In the following subsections we describe some of the most important process decisions to be taken in these phases, expressed in terms of GSBPM phases recognised in the big data lifecycle: Collect, Process, Analyse and Disseminate.

a. GSBPM 4 Collect: Acquisition/Recording

The process of collecting online-based enterprise characteristics (OBEC) data is a four-step process. It first requires identifying a list of companies for which data will be collected (target population), with basic attributes, like for instance company name. Then, a list of potential website addresses is built, for instance using responses from a website search engine for each company in the SBR or other administrative registers. In subprocess 3 a partial crawling data collection is done on potential website addresses (homepages), having a ranking mechanism to verify which website is likely to be the good one (thus searching for identification data on website). The final step would be, having chosen the "first-best" website for each enterprise/business, an extended crawling data collection done on websites, searching to get specific characteristics of enterprises/businesses.

There are two phases where collection of data occurs:

- **URLs retrieval:**

  Decisions on the search strategy are necessary during this process. Which search engines give the best results? What is their coverage in the subject country? Is one engine sufficient or is it necessary to use multiple search engines? Are they programmable? Can we use them for the possibly large amount of enterprises present in the SBR? How many queries do we issue per enterprise? What is a useful website (for example do we take Facebook and LinkedIn pages into account)? Are there additional data sources that can be used to find URLs, such as the data from the register of the national country code Top Level Domain (ccTLD)? Is there a dataset from commercial or semi-commercial companies scraping the national web that we can use? Do we perform - in addition to the search result - some scraping of the website to determine the correctness of the URL? Choices on the scope and frequency of the URL retrieval process are also required. Do we only search for missing URLs or are we also trying to verify URLs that are in the SBR taken from administrative sources?, etc.

- **Scraping websites:**

Decisions on the scraping process are necessary during this phase. Examples include decisions as to what depth to scrape the sites, whether to use any focus in the scraping (for example looking for contact pages or "about us" pages or other pages where meaningful information about the enterprise itself is expected). Decisions are also required about what parts of a page to be taken into consideration, the maximum number of links to follow, scraping frequency, and of course the more technical details of web scraping for which we refer to the preceding chapter on the application architecture and information architecture.

b. GSBPM 5 Process: Extraction, cleaning and annotation, integration, aggregation and representation, etc.

Data is processed across the various phases. We focus here on the interpretation of the scraped data:

- **Derive enterprise characteristics:**

In this phase the raw web data is cleaned first, both technically (removing html tags) and content-wise (removing stop words etc.). Decisions are taken on what parts of web pages are retained as inputs and what parts are discarded. Text parts that can be used as business identifiers are of special interest. Decisions on how to transform raw texts into structured content are necessary. Methods varying from a basic bag-of-words approach to methods preserving contextual information, such as doc2vec, word2vec, sentence2vec, or pre-trained networks for text analysis could be applied[10].

The text-to-variables conversion process involves deriving features, also known as feature extraction. Generally speaking, there are two possible approaches:

- *Deterministic approach*: decisions on rules used and their validity and maintenance over time are required.
- *Machine learning*, either supervised or unsupervised. In the case of supervised learning, obtaining or constructing a predictive training set is necessary. This well-known process is sketched in Figure 2. In the case of unsupervised learning, decisions on clustering algorithms to use and the mapping of clusters to the enterprise characteristics are required.

[10] An example is the work from Google on Bidirectional Transformers for Language Understanding (BERT).

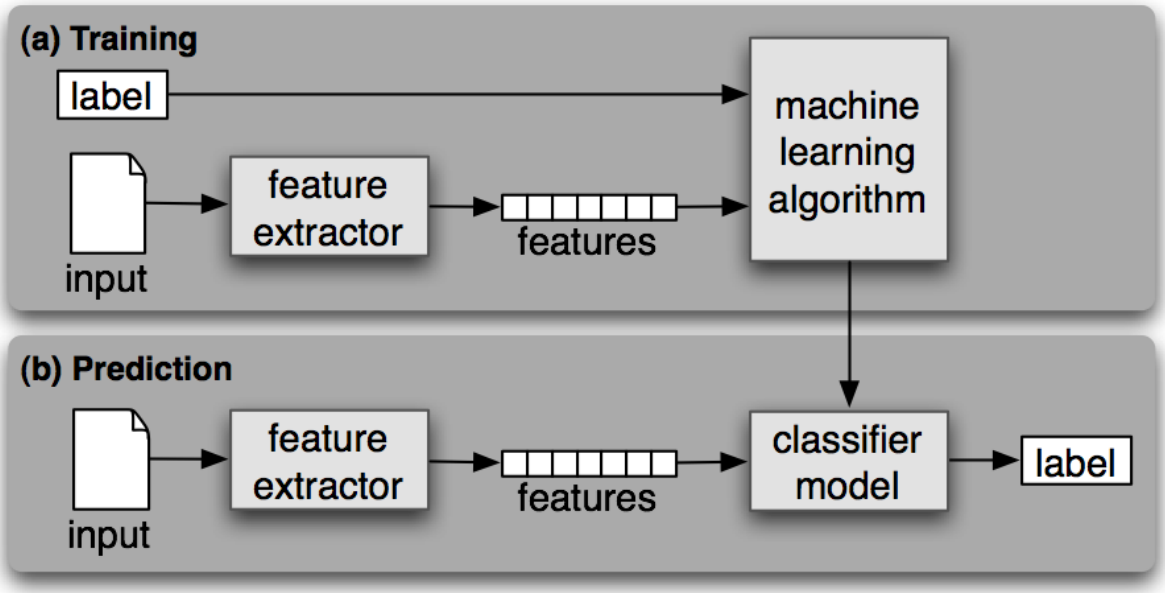


Figure 2: Supervised classification (from NLTK book chapter 6)

c. GSBPM 6 Analyse: Modelling and interpretation

The analysis of web data in combination with already existing data on enterprises in the SBR is crucial in many of the process steps:

- **Validate/reject candidate URLs:**

  Validating or rejecting candidate URLs needs careful analysis. Enterprise website and enterprise concepts are not the same. Mapping one to another is required. A website may contain information about multiple enterprises, or vice versa, an enterprise may own or publish on multiple websites. Which part(s) of the enterprise is covered by the website? Decisions on models to be used are required. This can vary from a relatively simple deterministic string distance function calculating the similarity of the name of the enterprise or other variables to the URL found, to a complex machine learning model taking many features into consideration. In the latter case, decisions on the features to use are necessary and is based on available SBR data and the expressiveness of the results of the search engine being used.

- **Derive enterprise characteristics:**

  The calculation of enterprise characteristics on a certain population based on scraped web data needs modelling and interpretation. Decisions on characteristic definitions are required, for example – on the micro level - decision criteria on whether a business is deemed to provide social media or ecommerce via its website. Also, note that there is almost always a subset of the SBR for which no URL could be found. Correcting for this on the macro level could be done by mapping the measured population to the target population.

d. GSBPM 7 Disseminate

Dissemination from enterprise characteristics derived from web data is not significantly different from the traditional dissemination processes, except that since these techniques are still less mature than traditional survey-based statistics, disclaimers and good explanations of the methods used could be added.

III. Reference architectures for online-based enterprise characteristics (OBEC) data (WPC, WPF)

This chapter is going to be completed in the next RMF version 2.0.

1. Application architecture

2. Information architecture

IV. Implementation requirements at ESS level

1. Functional production prototypes

The functional production prototypes have been produced by use cases. The first two of them are implementable in the statistical production process. The third and fourth are functional proof-of-concept prototypes and should be treated as such.

The detailed description of the functional production prototypes can be found in the Annexes section in this document.

2. Implementation requirements at national level

3. Set of recommendations for NSIs

V. Online-based enterprise characteristics (OBEC) in the context of traditional ICT survey

The annual "Survey on ICT Usage in Enterprises" ("ICT survey" for short) is carried out in EU member states under Eurostat regulations. It collects data on the usage of Information and Communication Technologies, the Internet, e-business and e-commerce in enterprises. In Italy, for instance, the target population covers all active enterprises with at least 10 employees. Enterprises with size 10 to 249 are sampled, whereas those with size 250 or more are all observed. The sampling design is a one-stage stratified simple random sampling, with strata defined by crossing economic activity (NACE), enterprise size and geographical region (NUTS1). The sample is drawn from ASIA, the Istat archive of about 4.5 million Italian active enterprises. In recent years, the planned sample size of the Italian ICT survey has been about 30,000 units and the response rate has been roughly 60%, yielding a respondent sample of nearly 20,000 enterprises (i.e. about 1% of the overall target population).

Some estimates produced by the ICT survey are related to characteristics of the websites used by enterprises to showcase their business, e.g. whether the website is used to provide e-commerce services, job advertisements and links to the company's accounts on social networks. Therefore, the idea of acquiring such information directly, that is by scraping and analyzing the content of enterprise websites, seems very natural and may offer a viable alternative to traditional data collection via questionnaires. Online-based enterprise characteristics (OBEC) obtained this way could allow shrinking the yearly ICT questionnaire or could be used to provide independent estimates to be compared or combined with survey-based ones. In particular, composite estimators obtained by suitably combining survey estimates and OBEC estimates have the potential to increase the quality of traditional ICT estimates by reducing their mean square error. Indeed, while the survey component of these estimators would provide protection against bias, the OBEC component would reduce the overall variance, thanks to the large coverage enabled by massive scraping. Going beyond the ICT survey, OBEC could also be routinely acquired and maintained as an additional information source to feed and keep the Business Register up-to-date.

As already sketched in previous sections, the process to access and exploit OBEC information typically involves the following macro-phases:

1) **URL Retrieval.** In this phase, website addresses are sought for the largest possible part of the population (be it the ICT survey target population or the reference population of the whole Business Register).

2) **Web Scraping.** In this phase, the websites whose URLs are known are accessed, their textual content is retrieved, cleaned, indexed and stored in a suitable platform.

3) **Feature Extraction.** In this phase, relevant text parts (words, word sequences or sentences) are identified within the scraped texts, with the aim of using them to predict the OBEC of interest (e.g. whether the enterprise performs e-commerce). To extract these features, it is of course necessary to exploit the subset of units for which the true values of interest (from the survey or the Business Register) and the web-scraped texts are both available.

4) **Modelling.** In this phase, a Machine Learning (ML) algorithm is trained to predict the OBEC of interest from the textual features extracted in the previous phase, using survey data and scraped data jointly as a training set.

5) **OBEC Prediction.** In this phase, the trained ML is used to predict the OBEC of interest for all the enterprises that were not observed by the survey (or by the Business Register) but whose websites were successfully scraped.

6) **Integration of OBEC information at macro and/or micro level.** In this phase, predicted OBEC values are integrated with traditional statistical information on enterprises. This can be accomplished at macro level and/or at micro level. In the first case, unit-level OBEC predictions are aggregated to yield estimates of population parameters (e.g. the proportion of enterprises offering e-commerce facilities). In the

second case, unit-level OBEC predictions are directly linked to the corresponding entities of the Business Register.

With respect to the **macro integration** of OBEC with existing statistical information on enterprises, in 2018 the Italian NSI started releasing experimental statistics on selected ICT variables (web ordering, job vacancies, links to social media). These experimental statistics have been made available at national level and for the following domains:

- classes of employees (4 modalities);
- classes of employees (4 modalities) crossed with NACE macro-sectors (4 modalities);
- NACE groups (26 modalities);
- NUTS2 (21 modalities).

Two different estimators based on OBEC values, $\tilde{y}_k$, have been devised: a *full model-based* estimator and a *composite* estimator. The corresponding estimates have been computed for both and compared with official ICT estimates obtained using traditional calibration estimators based on survey data, y_k. The characteristics of the proposed estimators are best understood taking into account the different subsets of the ICT target population, U :

- the subset U_1 of enterprises that own (or make use of) a website;
- the subset U_2 of enterprises that own (or make use of) a website, whose URL was retrieved and whose content was successfully scraped;
- the subset $U_1 - U_2$ of enterprises that own (or make use of) a website, whose URL was not retrieved or whose content was not successfully scraped;
- the respondent sample r of the ICT survey;
- the subset r_1 of enterprises that answered the ICT survey declaring to own (or make use of) a website, whose URL was retrieved and whose content was successfully scraped;
- the subset r_2 of enterprises that answered the ICT survey declaring to own (or make use of) a website, whose URL was not retrieved or whose content was not successfully scraped;

Traditional *calibration* estimators based on survey data:

$$\hat{Y}_{CAL} = \sum_{k \in r} y_k w_k$$

involve weights, w_k, that are as close as possible to direct weights (i.e. reciprocals of inclusion probabilities) but are adjusted, with the aim of reducing sampling variance and/or nonresponse bias in such a way that calibration estimates exactly match a given set of population totals. Among other constraints, calibration weights satisfy the following equation:

$$\sum_{k \in r} w_k = N_U$$

Therefore, the target population size estimated by traditional calibration estimators exactly matches its true value (typically derived from the Business Register).

The *full model-based* estimator built upon OBEC predictions:

$$\hat{Y}_{FMB} = \sum_{k \in U_2} \tilde{y}_k w'_k$$

involves pseudo-calibration weights, w'_k, fulfilling the following equation:

$$\sum_{k \in U_2} w'_k = \hat{N}_{U_1}$$

where $\hat{N}_{U_1}$ is the (unbiased) estimate of the size of U_1 derived from the ICT survey. This way, potential bias arising from the failure of the URL retrieval and/or scraping phase is hopefully reduced and the "reached" subpopulation U_2 is made as representative as possible of the interest subpopulation U_1. Note that, even though $\hat{Y}_{FMB}$ actually exploits some information coming from the traditional ICT survey, it does not directly involve survey values y_k.

The *composite* estimator does involve OBEC and survey based values simultaneously:

$$\hat{Y}_{COM} = \sum_{k \in U_2} \tilde{y}_k + \sum_{k \in r_1} (\tilde{y}_k - y_k) w''_k + \sum_{k \in r_2} y_k w'''_k$$

and two sets of pseudo-calibration weights, w''_k and w'''_k, such that:

$$\begin{cases} \sum_{k \in r_1} w''_k = N_{U_2} \\ \sum_{k \in r_2} w'''_k = \hat{N}_{U_1 - U_2} = \hat{N}_{U_1} - N_{U_2} \end{cases}$$

The formula for $\hat{Y}_{COM}$ entails three contributions:

1) the simple (i.e. unweighted) sum of OBEC predictions in the subpopulation U_2;

2) an adjustment term accounting for the differences between OBEC predictions and survey values when both are available, that is for units belonging to r_1. Such differences are expanded through weights w''_k whose sum matches the size of the relevant subpopulation U_2;

3) the weighted sum of survey values for units belonging to r_2, namely enterprises which did respond to the survey but whose website was not reached. The employed weights w'''_k are such that their sum matches the estimated size of the relevant subpopulation $U^1 - U^2$.

OBEC based estimates produced by means of $\hat{Y}_{FMB}$ and $\hat{Y}_{COM}$ have been compared to traditional ICT estimates $\hat{Y}_{CAL}$ for the whole population and for domains. Empirically, it seems

that the three sets of estimates are not incoherent. For instance, both OBEC estimates of the proportion of enterprises with web ordering facilities fall within the limits of survey based 95% confidence intervals. The same happens for many domains, though OBEC estimates unsurprisingly show larger deviations from traditional estimates for smaller domains. Overall, adopting the traditional survey estimator as a benchmark, the combined estimator exhibits a slightly better performance than the full model-based one. However, the full model-based estimator would be the only possible choice in case the decision is made to remove the "website section" from the ICT questionnaire.

With respect to the **micro integration**, the main idea is to link unit-level OBEC predictions to the corresponding entities of the Business Register. This linkage operation is expected to be straightforward, as the association of OBEC information to a given unit of the Business Register would typically be already accomplished in the 'URL retrieval' macro-phase. In fact, URLs are searched starting from a set of key variables available in the Business Register, such as the ID and the name of the enterprise. No matter how complex the URL retrieval procedure could be, it ends up either without any URL found, or with a discovered URL that is thereby natively associated to the enterprise ID. Such a binding between the enterprise ID and the retrieved URL is, of course, maintained also in the subsequent macro-phases of the process. As a consequence, an unambiguous link between the enterprise ID and the output information of the 'web scraping' and 'OBEC prediction' macro-phases is also automatically gained.

VI. Data quality and metadata

Although there are several different frameworks for data quality, there are some fundamental indicators that can be used to describe quality in the web scraping process. According to the work done by ESSnet Big Data I WP8 Methodology[11] there are five main group of measures that can be used to assess data quality. They include: coverage, accuracy and selectivity; measurement errors; comparability over time; ease of linking; processing errors. Later, in the Quality Guidelines for the Usage of Big Data prepared by ESSnet Big Data II WP-K Methodology, web scraping was described as a distinct big data class. This class can be used to acquire data from web portals (job vacancies, enterprise characteristics, comments/news) as well as Application Programming Interfaces - API's (Twitter, Google or Bing searches etc.)[12]. This class has been mostly described in the throughput phase, according to the big data three phases framework – input, throughput and output[13].

[11] ESSnet Big Data WP-8 Methodology https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WP8_Methodology1, as of 5th of August 2019

[12] ESSnet Big Data II WP-K Quality Guidelines for the Usage of Big Data (Draft in Process)

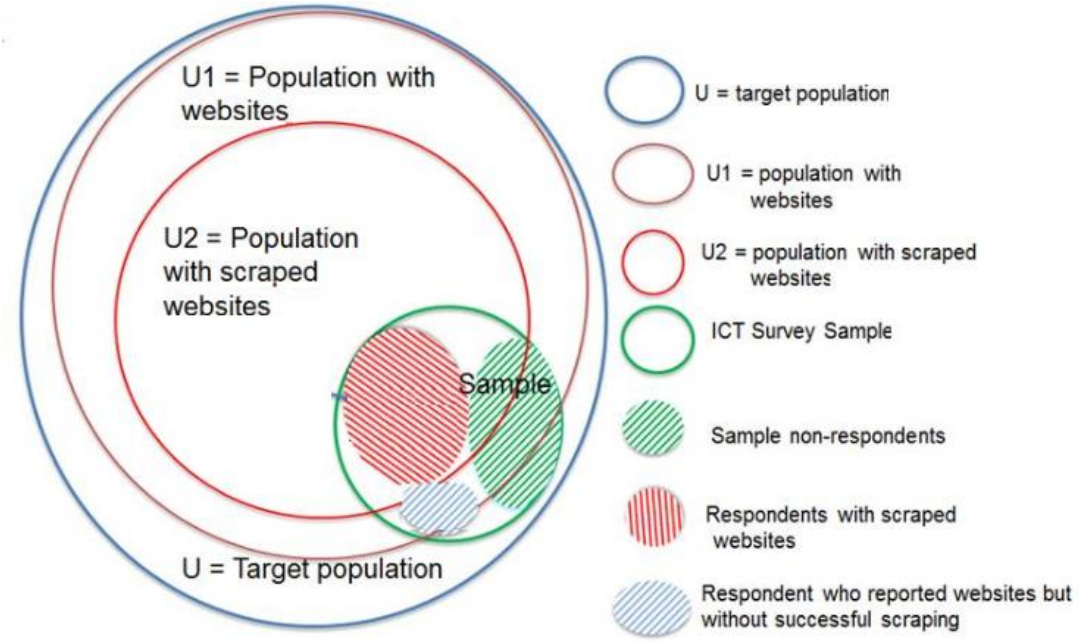
[13] A Suggested Framework for the Quality of Big Data, UNECE, 2015, https://statswiki.unece.org/display/bigdata, as of 5th of August 2019

1. Quality aspects of web scraped enterprise characteristics data

a. Coverage, accuracy and selectivity

Coverage is one of the most important factors that allows us to prepare big data surveys that can be measured in terms of representativeness. One of the most important issues is being sure that the selected population fits to the target population. This is illustrated in Figure 1.

Figure 1. Scraped vs. target population



Source: ESSnet Big Data WP-8 Methodology
https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WP8_Methodology1, as of 5th of August 2019

In Figure 1 we can see that the sample from traditional surveys may not be available in the web scraping process, as there are enterprises that do not have websites.

Due to the fact that the sample of web data is unknown in many cases, coverage is a relevant issue we have to tackle during web scraping. According to the specification of the dataset, it includes various aspects of coverage. For example, web scraping Twitter data leads to the conclusion that the data are not representative because it is not commonly used and there is usually under-coverage of different groups of individuals. When web scraping websites, we should be aware of the fact that not all enterprises are present in the web. Especially, small enterprises and self-employed persons that may not be present on the web or social media. Therefore, there is still a problem of over-coverage of large enterprises in contrast to small enterprises.

For enterprise web sites it is essential to obtain the corresponding URL of all companies in the Business Register. However, not all businesses have a web site and some types of businesses have a higher change of having a web site. This affects coverage considerably. By specifically searching for the website of a company via a search engine, an attempt is made to solve this issue on a case by case basis. Coverage may be measured at three different levels: micro-level,

aggregation by industrial sector, and advertising channels. All three were found to be selective in some aspects (Swier et al. 2018). [*WP-K, ESSnet Big Data II WP-K Quality Guidelines for the Usage of Big Data*].

b. Measurement error

The concepts measured (or derived) by scraping web pages may not be identical to the ones required by the NSI. This may cause measurement errors. An example of this is found in online vacancies where the classification of jobs was extremely difficult because different web sites used different standards; producing errors. Similar errors occur for important concepts such as occupation and economic activity. For these reasons, NSIs need to consider what the best use of their time and resources is. NSIs should consider options that will minimize the amount of data handling required to reduce these errors. [*WP-K, ESSnet Big Data II WP-K Quality Guidelines for the Usage of Big Data*].

c. Comparability over time

The internet changes continuously. Both the content and the structure of a web page may change. On the one hand, web scraping may become increasingly challenging as the internet evolves, as more and more data may be encoded in forms that are harder to extract – audio or video files, for example, or increasing use of interactive or user-specific content. It is therefore easy to imagine that websites of interest, such as those of some enterprises in the creative industries, for instance, become much more challenging to extract information from. Such web scraping may therefore need to be increasingly carried out by specialists inside or outside of NSIs, rather than by only data scientists. On the other hand, when the content of web pages is continuously updated it is a challenge to assure all data is scraped from a web page. For some web sites this may require a daily visit and puts a burden on data management (Swier et al., 2018). [*WP-K, ESSnet Big Data II WP-K Quality Guidelines for the Usage of Big Data*].

d. Ease of linking

Linking a web site and a company in the business register is essential for web scraping. This is often not known for all businesses. With search engines a list of websites can be retrieved and related to a given enterprise. Usually this list is obtained by querying a search engine on the web using the name of the enterprise as a search term. The underlying assumption is that, if an enterprise has an official website, this should be found within the results provided by the search engine. By comparing any company specific data displayed on the web page, such as a Chamber of Commerce number, it is possible to check this relation. Using official documents of a company, such as an annual report, is another way of obtaining the appropriate URL of the website. For online vacancies, several countries have explored the matching of online job ads with their own Job Vacancy Statistics micro data or Business Register data. The results have been somewhat mixed and vary from good linkage, when a common identifier was present, to bad, when probabilistic linkage was used (Swier et al., 2018). [*WP-K, ESSnet Big Data II WP-K Quality Guidelines for the Usage of Big Data*].

A typical approach to data linkage is presented in Table 1.

Table 1. Linking web scraped data with enterprise

Specification	**Description**
Type of data	URLS of Enterprises
Entity	Company
Additional source	Business register
Entity identifier	Register number, name, address (depends)
Method of linking	Machine learning

Source: ESSnet Big Data WP-8 Methodology
https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WP8_Methodology1, as of 5th of August 2019

Both, generic scraping of whole websites and specific scraping of predefined content can be used to collect information on websites.

e. Processing errors

As mentioned above, the biggest issue is the continuous change of the Internet. It means that we should monitor changes on the Internet and modify the web scraping software if necessary.

For enterprise characteristics a processing pipeline was created. Setting up a pipeline assures processing is comparable over time. Because texts were processed, the final results were highly affected by the various choices of text processing made. [*WP-K, ESSnet Big Data II WP-K Quality Guidelines for the Usage of Big Data*].

f. Modelling errors

The non-representativity of web scraped data introduces bias in the estimates. Some of the pilots in the ESSnet have highlighted the issue of bias – for example, they were more likely to identify websites for businesses conducting e-commerce, and e-commerce statistics purely based on data scraped from the websites found and will therefore be biased without adjustment. They also found that some websites – for example, those that make heavy use of JavaScript – are harder to scrape than others, and this may also introduce bias. A simpler source of bias, for some statistics, will simply be that some businesses are less likely to have a website than others.

A key challenge will be to understand these biases for different usecases and work out how to adjust for them so that scraped data can be used for estimates qualifying as official statistics. This is likely to involve methodological work around combining web-scraped data and survey or administrative data (Stateva et al., 2017b).

In a number of projects, machine learning methods were applied and compared with the more commonly used deterministic approaches in official statistics. An important lesson was that similar results could be achieved with different methods and that some findings even converged (Stateva et al., 2018b). One of the factors that influenced the choice between deterministic and machine learning methods is the complexity of the relationship between the input variables (or

the features) derived from the input data and the statistical target variables. If this relationship is fairly straightforward a deterministic approach seems obvious. But if this relationship is complex, unknown or difficult to model in a deterministic algorithm, which might well be the case when working with web data, a machine learning approach might perform better. For the latter, generalisation and overfitting were found to be import quality issues. *[WP-K, ESSnet Big Data II WP-K Quality Guidelines for the Usage of Big Data]*.

With respect to the work done by ESSnet Big Data I WP2 team, statistical models were applied in the analysis layer. These models were needed, since it was not possible to go individually through every scraped website to check if the website contained information on job vacancies, if the enterprise was engaged in e-commerce or if there existed information about social media presence. Therefore, models were needed to predict the dependent variables from scraped information and correctly classify the enterprise.

The members of WP2 describe the machine learning approaches as well as deterministic techniques that were applied: we speak about machine learning approaches in scraping for official statistics when algorithms or models are derived from a set of training data which is supposed to be reasonably representative for the problem at hand. The parameters of the model are usually tuned with a validation set before measuring its performance on a so-called test set with known characteristics. Finally, the model is then applied to other sets of data of which we do not know anything but for which we suppose the model performs well, in order to produce statistics.

We speak about deterministic approaches in scraping for official statistics when algorithms are designed from a set of rules with known characteristics of the sites and data in mind. Put in a different way, the knowledge of an expert is used to design an algorithm to process and interpret input data from web and other sources into statistical target variables. We call this method deterministic because the algorithm applied to the same data will always have the same (deterministic) result, where the result of a machine learning approach heavily depends on the training set being used. It is clear, that especially in the machine learning case, the quality of the model depends on the quality of the training data and on the test data to make validation possible. This is also described in the report: one thing to be noted here is that in machine learning approaches the availability of training data of sufficient quality is essential. This happens to be a challenge in many cases. In some of the pilots this training data is available or can be derived from earlier surveys. This might be true the moment when a machine learning approach is introduced in official statistics to (partly) replace a traditional statistical process, however in the long run, survey-based training data might become a rarity and other means have to be found to (re)train machine learning models. Obviously, deterministic approaches do not have this challenge, but have other pitfalls. [WP2 Deliverables].

2. Quality report for web scraping of enterprise characteristics

This sub-section is going to be completed in the next RMF version 2.0

VII. Annexes

Annex 1: Functional production prototype **URLs Inventory of enterprises**

Overview (brief description)
The use case *URLs Inventory of enterprises* is dedicated to determining whether an enterprise has a website. The goal is to generate a URLs Inventory for enterprises with 10 or more employees (10+ employees) consisting of enterprise ID, name and URL. The result can be used to retrieve information from the enterprise websites on variables in the ICT usage in enterprise survey, new variables not included in the ICT survey and validation of the SBR or NACE classification. The statistical indicator - *Rate of enterprises having websites* has been defined. This indicator could replace the question from the ICT survey "Does your enterprise have a website?" or it could be used to validate survey responses.
Data Sources (used data sources)
The data sources used are the following: • national statistical business register • ICT survey • internet sources (e.g. search engines, yellow pages) • administrative registers
Description (methods used, SW used, procedure adopted, decisions, etc....)
1. Methods used The methods used are as follows: • querying one or more search engines (preferably via an API) with different variables from the SBR to obtain a list of suggested enterprise URLs; • scraping the suggested URLs or using variables from the search results (such as 'snippets') to obtain text information on enterprises; • text analysing/matching the obtained information to SBR variables; • scoring the suggested URLs from the matched information; • machine learning to choose the URL with highest confidence score for the enterprise (where no URL is chosen if all the scores are below a threshold). 2. Software used The following software was used: • search API services (Google, Bing) • ISTAT Java software, Solr, R

- CBS S4SGoogleSearch Node
- CBS ML Python software (in use, but not yet published)
- BNSI PHP, MySQL

The software used is available on the WPC GitHub:

https://github.com/EnterpriseCharacteristicsESSnetBigData?tab=repositories

3. Procedures adopted

a. Java, Solr, R

The procedure contains several steps.

Step 1

A custom Java program (UrlSearcher) takes in two input files. One containing the list of the firm names and one containing the corresponding list of firm IDs. For each enterprise the program queries a search engine (we are using Bing) and retrieves the list of the first 10 URLs provided by the search engine. These URLs are then stored in a file so we will have one file for each firm. Finally, the program reads each produced file and creates a seed file. The seed files are tab delimited text files where each row is composed as follows:

url_retrieved + TAB + firm_Id + TAB + position_of_the_url

e.g. www.mywebsite.com/aboutus 111 3

Step 2

A custom Java program (UrlCrawler) takes in three input files:

1) the seed file just produced;

2) a blacklist of domains to avoid (usually directory domains);

3) a configuration file.

For each row of the seed file (if the domain of the URL is not in the list of domains to avoid) the program tries to acquire the HTML page. From each acquired HTML page the program selects the textual content of the fields we are interested in and writes a line in a CSV file. Each row of the CSV file contains the following variables separated by tab characters:

id
url
imgsrc
imgalt
links
ahref
aalt
inputvalue
inputname
metatagDescription
metatagKeywords
firmId

companyWebsite
link_position
title
text_of_the_pagebody

Step 3

The CSV file is imported into the storage platform (we are using Solr).

Step 4

A custom java program (UrlScorer) takes in the following as input:

1) a CSV file where each row contains information about a specific firm (e.g. ID, name, VAT code, ZIP code, telephone number);

2) a reference to the storage platform.

For each row of the file (that is for each firm) the program extracts all the documents previously stored for that firm from the storage platform. For each document the program calculates a score vector and assigns a score on the basis of predefined conditions (e.g. in the text of the document we can find the VAT code or the telephone number) and writes a row in a CSV file with the following structure:

firmId + TAB +
linkPosition + TAB +
url + TAB +
scoreVector + TAB
score

The score vector is a vector of features with the following structure, for example: 2111111 where the digits read from the left to right indicate:

1) telephone number found in the text (2: found, 1: not found);
2) URL in a “simple” form like www.firmname.com (1: yes, 0: no);
3) link position (9: 1st position, 8: 2nd position, 7: 3rd position, …, 0: 10th position);
4) VAT code found in the text (1: yes, 0: no);
5) municipality name found in the text (1: yes, 0: no);
6) province abbreviation found in the text (1: yes, 0: no);
7) zip code found in the text (1: yes, 0: no);

The score is an integer from 0 to 1000.

Step 5

The CSV file obtained in step 4 is the input for fitting the machine learning model that tries to predict the correct URL for each firm.

In order to be able to accomplish this task it is necessary to provide a training set in which both the score vector and the indication of whether the link corresponds to the correct URL are available for each link.

A test set is created using a custom Java program (UrlMatchTableGenerator) that merges the CSV file from step 4 with a list of correct sites.

This list of correct sites can be obtained in different ways:

- by using data produced by the EU "ICT usage in Enterprises" (one of the questions regards the availability of a website, and in the positive case, the related URL is indicated);
- by acquiring data from third parties;
- by manually inspecting on the web.

Different models can be fitted on this subset of links and their performance in terms of accuracy and other quality indicators can be evaluated.

In the R script "models_estimation.R" three different learners are fitted (logistic, random forest and neural networks), and all three are saved as "name_learner.RData".

On the basis of performance one of the learners is chosen.

In the current version of the script the logistic model is chosen and it is used to verify the distribution of true and false positives in 10 different classes of the predicted score of being the correct link. This distribution is output in the file "res_logistico.txt", that is the basis for the determination of the thresholds that will be used in the next in order to assign the status of "link" / "non link" / "to be decided". For instance:

score_class	true	false	classification_error	group
[0.0312,0.124]		440	3239	0.880402283	1
(0.124,0.254]	628	3312	0.840609137	2
(0.254,0.369]	859	2569	0.749416569	3
(0.369,0.507]	1473	2290	0.608557002	4
(0.507,0.573]	1895	1555	0.450724638	5
(0.573,0.725]	3038	830	0.214581179	6
(0.725,0.862]	2958	561	0.1594202907	
(0.862,0.921]	3188	428	0.118362832	8
(0.921,0.936]	1397	141	0.091677503	9
(0.936,0.943]	5222	480	0.084180989	10

Step 6

In this final step the model fitted in the previous step is applied to all the links not belonging to the dataset used to fit the model.

This step is performed by using the script "apply_logistic.R". Inputs to this script are the following:

- the dataset of all the links not used to fit the model;
- the estimated model ("logisticModel.RData").

Pre-processing of the input consists in considering, for each enterprise, only the link with the highest score. The model is applied to this link and the predicted score is obtained. On

the basis of this value, and of thresholds for the assignment of the status, to each link a value "link", "non link" or "crowd" is assigned ("crowd" means that the status has to be decided by manual inspection, preferably by making use of a crowdsourcing platform).

Inside the script the thresholds for the assignment of the status must be inserted, on the basis of the distribution of the predicted scores:

```
#-----------------------------------------------------------------------
# ATTRIBUTION OF LINK/NONLINK/CROWD
# the values reported here are based on the results obtained by the fitting of the logistic
# (see the file "results.csv" output of the models_etimation script)
urls$link <- ifelse(urls$predictedScore > 0.573,1,0)
urls$nonlink <- ifelse(urls$predictedScore <= 0.507,1,0)
urls$crowd <- ifelse(urls$predictedScore > 0.507
                & urls$predictedScore <= 0.573,1,0)
#-----------------------------------------------------------------------
```

Values "0.507" and "0.573" are such that all first for classes are defined as "non links", while the last five are defined as "links"; the fourth class is destined for manual inspection.

The procedures for URLs inventory at Statistics Netherland are conceptually slightly different from the above steps. One of the differences is that the search results from the Google search step, including the snippet, are used to find the right URL for an enterprise. A detailed description of this method can be found in the research paper Van Delden, Windmeijer and ten Bosch, Finding enterprise websites, The European Establishment Statistics Workshop, Sep 24-27 2019, Bilbao, Spain, https://webgate.ec.europa.eu/fpfis/mwikis/essnetbigdata/index.php/WPC_Experimental_statistics.

<u>PHP, MySQL</u>

1) Get enterprises from the ICT population with the following characteristics: IDs, names, URLs, email addresses and other characteristics form the SBR in CSV file.
2) Create a MySQL database table with IDs, names, URLs, email addresses and other characteristics fields from the CSV file of the enterprises.
3) Upload the data from CSV file to the database table.
4) Get Google search API key.
5) Configure the information in the conf.php file.
6) Add necessary database table fields according to the information in the conf.php file where the output information from the project's software will be saved.
7) Upload the known and verified URLs of the enterprises from the previous iterations to the database table for sustainability results.

8) Check if the known URLs from SBR have working websites, generate URLs from the email address domains of the enterprises (excluding the popular emails domains: Yahoo, Gmail, etc. form the list in the conf.php file.) and check the generated URLs for working web sites with geturl.php script.
9) Check the known and verified URLs of the enterprises from the previous iterations for working websites with checkoldurl.php script since the changes have likely occurred from the previous period.
10) Run google_search.php script to get up to 10 suggested URLs of the enterprises from the results of running Google search API with the enterprises names.
11) Make a list with enterprise IDs and a corresponding list with enterprise names from a MySQL database containing enterprises' data for the ICT survey target population.
12) (Optional) Run ISTAT software UrlSearcher.jar either stands alone or through UrlSearcher.php script to get to up to 10 suggested URLs of the enterprises form the results of running Bing search API with the enterprise names.
13) Run list.php script to make manual verification of the correct URLs of the enterprises from the checked URLs of the SBR and previous iterations and the suggested URLs from the Google (and Bing) search APIs. After running this step, you have a list of enterprises with known URLs in the DB table.
14) Run the info.php script to see statistics from the above executed steps for URL retrieval at regional level and NACE categories.

Conclusions (lessons learned, open issues, obtained results, ...)

The obtained results are promising and add information for the enterprise URLs available in the national SBRs. This process was tested several times in different EU countries with success. Therefore, it is suitable for integration into real statistical production.
The process of URL retrieval needs to be repeated regularly for updating and maintaining the URLs inventory and to check validity of URLs. The frequency could be at least once a year.
Depending on the data update ambition of each NSO, the volume scale does not necessarily need to be the complete enterprise list, one could opt for a rotating sample.
The stability of internet as a data source for official statistics also needs consideration because the policies of search engines vary. For example, Google Search API is limited to 10,000 paid queries per day. Therefore, it is more time and resource consuming for larger EU countries.
In general, Python, Java, PHP, Node and R are the main web scraping languages. The experienced gained shows that the most convenient programming languages, used for URL retrieval, depend on in-house competence.

Annex 2: Functional production prototype **Variables in the ICT usage in enterprise survey**

Overview (brief description)
The use case "Variable in the ICT usage in enterprise survey". The goal of this use case is to answer the following questions: • whether an enterprise performs e-commerce or not? • whether an enterprise is present on social media? • whether an enterprise manages job vacancies on its website? • whether a website has some specific functionality (price lists, description of goods and services etc.)? • whether an enterprise gathers any information regarding user behavior? • how frequently the website is changed/updated? To answer the aforementioned questions five different indicators have been prepared. 1. E-commerce - rate of enterprises engaged in web sales on their website. 2. Social media presence - rate of enterprises that are present on social media. 3. Rate of enterprises using Twitter for a specific purpose: a. recruitment b. developing the enterprise's image or marketing products (e.g. advertising or launching products) c. others – any tweet that does not fit into the other two categories 4. Rate of enterprises having specific features on the website (e.g. list of products, list of services offered, shopping cart). 5. Rate of enterprises working on upcoming/new phenomena, specifically the use of artificial intelligence (AI) and machine learning (ML).
Data Sources (used data sources)
• Websites of every enterprise having a website and employing more than 10 persons. • Business register data (ID, address, website)
Description (methods used, SW used, procedure adopted, decisions, etc....)
4. Methods used ○ web scraping ○ text mining ○ supervised machine learning 5. Software used ○ Python with pandas and sklearn ○ Social media presence software ○ Alternatively R-Code using RSelenium and rvest, including README and vignette.

The software used is available on the WPC GitHub:

https://github.com/EnterpriseCharacteristicsESSnetBigData?tab=repositories

6. Procedures adopted

There are two sets of methods used to provide output data for indicators:

1) web scraping / text mining – used for indicators 1, 2, 4 and 5;
2) web scraping with API / text mining / supervised machine learning – used for indicator 3.

Set 2 is more vulnerable to changes and less reliable than set 1. This is because machine learning is sensitive to differences between the training dataset and the data it is being applied to.

A description of how to use the prototype, written in Python, is included below.

(1) Collecting social media links

Use the SocialMediaPresence_Dev application to collect social media links for each enterprise. The input for this application is the list of URLs, for example:

maslankowski.pl

http://stat.gov.pl

www.ug.edu.pl

The output will be a CSV file with the links to the following social media: Facebook, Twitter, YouTube, LinkedIn, Instagram and Google Plus

grouped by enterprise.

(2) Collecting Tweets for training and testing purposes

The goal of this step is to collect tweets with two applications to prepare the training and testing dataset:

Step2a_CollectingTweetsByKeyword

Step2b_CollectingTweetsByEnterprise

The result of the application is a CSV file having all tweets associated with a specific keyword or from a specific enterprise. The goal of them is to group tweets by keyword or enterprise to make the creation of a training dataset easier.

To each tweet one of the following categories should be assigned according to the ICT in enterprises survey:

- commercial
- marketing
- enterprise image
- recruitment
- others

(3) Training and testing a dataset

The aim of the application Step3_PurposeOfSocialMediaPresence

is to validate the training dataset.

In this step the validation of training dataset is tested. Our goal is to achieve an accuracy higher than 80%.

(4) Apply the data to check enterprise main purpose of social media presence

The final application, Step4_PurposeOfSocialMediaPresence, is used to prepare experimental statistics on the use of social media in enterprises.

The result of this application is information on the number of tweets grouped in five categories:

- commercial
- marketing
- enterprise image
- recruitment
- others

Conclusions (lessons learned, open issues, obtained results, …)

There are several different aspects of this use case. Firstly, we have to be aware of the fact that the target population does not include the whole population of the websites of enterprises having 10 employees or more. Secondly, some websites may be incorrectly identified as belonging to the enterprise. Nevertheless, the proportion of such errors is not significant and we can draw conclusions based on the data from websites.

As with a traditional survey it is also important to gather data from websites in the same reference period. For consistency with the ICT in enterprises survey this period would need to be February to April every year.

It is recommended to gather the same data several times in the reference periods. A case study for Poland shows that the value of the same indicator may change even 1.86% in two weeks (e.g., social media presence). This was a result of the fact that response from all websites are not received every time.

It is also very important to set the maximum allowable response time for a website as they may not always return a response in a timely manner.

Based on the results of our analysis it is highly recommended to decrease the number of aforementioned categories if the training dataset does not contain enough observations or if there is a category imbalance in the training dataset (e.g. 1 thousand for commercial, 6 thousand for recruitment, 15 thousand for others).

Annex 3: Proof-of-concept prototype **Data driven discovery of emergent enterprise classifications**

Overview (brief description)
Given a huge text corpus obtained by a massive web scraping activity on enterprise websites, natural language processing techniques (e.g. word embeddings, latent Dirichlet allocation) will be used to: 1) explore the semantic content of the corpus; 2) identify meaningful semantic and syntactic patterns; 3) use the discovered patterns to define data driven enterprise classification; 4) associate each enterprise with the most appropriate class of the discovered classifications. The expected outcome is one or more new enterprise classifications and the corresponding distributions of the scraped enterprises.
Data Sources (used data sources)
Administrative data ICT survey data Business register data Enterprises websites
Description (methods used, SW used, procedure adopted, decisions, etc....)
From a conceptual point of view, the work done in this proof of concept prototype can be divided into two main phases: 1) Data acquisition 2) Data analysis **Phase 1 – Data acquisition** The objective of this proof of concept prototype is to discover emergent enterprise classifications in a fully data driven fashion. In particular, enterprise websites will be used as a data source. Since the number of enterprises to consider is quite high all the steps have to be automated as much as possible. In order to be able to crawl and scrape content from enterprise websites it is necessary to have the full list of websites of interest. In our case this requirement was not satisfied so we needed to procure the full list with a dedicated URL retrieval procedure. Once we obtained the URLs of interest we were able to scrape them by using a custom Java program that produces text files containing the textual content of the websites. In the following a brief description of these two steps (URL retrieval and URL web scraping) is provided.

URL retrieval

The first step was to merge all the already available URLs coming from different traditional sources, in particular we merged data from:

- ICT survey (survey data)
- Consodata (administrative data)
- Asia (business register).

The remaining part of the URLs considered in this proof of concept (the URL list obtained after the merge operation was not exhaustive) was obtained by using an automated procedure. The main steps of the procedure are as follows.

1) To acquire information (business name, address, telephone number, VAT code, etc.) about each enterprise of interest from existing sources (Business Register plus other archives)

2) For each enterprise in the target population:

- introduce the business name into a search engine
- obtain the list of the first k resulting web pages
- scrape the content of the first k resulting web pages
- For each one of these results, calculate the value of binary indicators. For instance:
    - the URL contains the business name (Yes/No);
    - the scraped website contains geographical information coincident with already available in the Register (Yes/No);
    - the scraped website contains the same fiscal code in the Register (Yes/No);
    - the scraped website contains the same telephone number in the Register (Yes/No);

3) Compute a score on the basis of the values of the above indicators.

4) On a subset of enterprises for which the URL is known (training set), model the relation between the binary indicators plus the score, and the success/failure of the found URL.

5) Apply the model to the list of enterprises for which the URL is not known in order to decide if the found URL is acceptable or not.

URL Web Scraping

We scraped the websites of each enterprise in the target population by using a custom Java program that:

- tries to acquire a predefined number of webpages for each website
- extracts the textual content from each acquired webpage
- writes a line containing the scraped text in a CSV file

The produced CSV file represents the input for the data analysis phase of the proof of concept.

For the URL retrieval step we used the following programs:

UrlSearcher (https://github.com/SummaIstat/UrlSearcher)

RootJuice (https://github.com/SummaIstat/RootJuice)

SolrTSVImporter (https://github.com/SummaIstat/SolrTSVImporter)

Apache Solr (https://lucene.apache.org/solr/)

UrlScorer (https://github.com/SummaIstat/UrlScorer)

UrlMatchTableGenerator (https://github.com/SummaIstat/UrlMatchTableGenerator)

Custom R scripts

For the URL retrieval step we used RootJuice (https://github.com/SummaIstat/RootJuice).

Phase 2 – Data analysis

For our enterprise classification system we use a primarily data-driven approach. It uses the ability of Word embedding (WE) models to capture word semantics, and ML algorithms to group entities with similar characters.

The procedure implemented consists of two steps. In the first step we construct groups of words that represent pseudo-topic. In the second step the websites of the companies are profiled through a barcode (technically speaking this "barcode" is a bar plot, we decided to call it “barcode” because it can be considered a sort of unique identifier) which is a function of the frequency with which the words of the site belong to the various pseudo-topics. Subsequently the sites are clustered according to their barcodes. The classification of enterprises must emerge from this clustering. Each cluster of the emerging classification is assigned a label that depends on the barcode of the archetype or centroid of each cluster.

Search of Pseudo-topics

We train a word embedding model on the corpus consisting of all the text generated by the web-scraping procedure. The algorithm chosen is word2vec and the output of this algorithm is a mapping between the dictionary of words in the corpus and a vector space of 200 dimensions. The advantage of this representation is that the geometric relationships between the word vectors contain the semantic information between them.

The words represented as points in the word embedding space are therefore clustered into k clusters, but only after dimensionality reduction has been performed with t-SNE (t-Distributed Stochastic Neighbor Embedding); it is particularly efficient for the word embedding models as it maintains the local structures. After the t-SNE dimensionality reduction the pseudo-topics are generated by the k-means clustering algorithm. The labels of each pseudo-topic are therefore assigned through the clerical inspection of the most frequent words.

The website barcode profiles

For each website relating to an enterprise of interest, the barcode is generated containing a vector with a size equal to the number of pseudo-topics. In each position of the vector we report the normalized frequency of the words related to the pseudo-topic.

With the aim of grouping enterprise websites that deal with similar topics (represented by parallel vectors of the vector space), we create a second cluster using the k-means algorithm. The result of this step represents the output of the entire procedure.

Now we are two possibilities:

a. A visual inspection of the enterprises contained in each cluster with the aim to infer a label of the classification based on the presence of common characteristics.
b. Compute the centroid of each cluster and analyze the corresponding barcode with the aim to infer a label of the classification based on the most frequent and "meaningful" pseudo-topics available.

The very last step is to inspect the obtained classification and evaluate the quality of the results. It could be possible that the results obtained during the first round of the procedure don't fit the expectations, in this case a possibility is to run the procedure again with different parameters and hyper-parameters for the algorithms on the basis of previous experience and personal intuition.

To achieve this we developed custom python programs using several data science libraries (gensim, sklearn, pandas). For the word embedding part of the procedure the word2vec C program was used. In order to facilitate subsequent code sharing among other participants, we decided to use Jupyter notebook as our IDE.

Conclusions (lessons learned, open issues, obtained results, ...)

Obtained results

The results obtained so far didn't match our expectations for the following reasons:

- Several pseudo-topics were totally non relevant (e.g. stop-words, list of cities, numbers, etc.);
- Some pseudo-topics are too generic in the sense that too many different concepts are put together;
- Some clusters contained different pseudo-topics that are difficult to synthesize in just one concept.

On the other hand, a small number of clusters are promising because it was possible to easily identify emergent enterprise classification.

Lessons learned

We observed that a full data-driven approach does not initially provide high quality results, moreover our specific process is not fully data-driven due to the fact that we manually assigned labels to the obtained pseudo-topics.

Despite previous experience in the field it is not easy to identify the correct parameters and hyper-parameters for the used algorithms.

Open issues

There are still some actions that could be put in place in order to improve the quality of the obtained results; in particular we are thinking about the following experiments:

- Increase the number of words considered in the dictionary with the aim to catch emergent (but not widespread) phenomena. The related disadvantage in this case will be the amount of noise incorporated (at the moment we are cutting off less frequent words).
- The usage of human-made pseudo-topics will probably increase the quality of the second clustering (the final one). The disadvantage could be the lack of ability in recognising new emergent and not well known phenomena, it seems obvious that an appropriate tradeoff has to be found.
- The disaggregation of the most promising pseudo-topics could lead to better final pseudo-topics in terms of specificity (e.g. we could disaggregate a first level cluster made by names of products in order to obtain a good number of specific second level cluster organized by topic).
- In order to reduce the amount of noise present in the scraped data it could be an interesting idea to consider just a subset of the scraped web-pages, in particular it could be interesting to consider just the “about us” pages.

Future work

In the near future we will surely work on the previously described open issues, in addition, some good points to be investigated are:

- To change the algorithm used for the topic identification (e.g. LDA instead of word embeddings/k-means)
- To find a better configuration of parameters and hyper-parameters, for example:
    - increase the number of dimensions in t-SNE with the aim of avoiding the compression of the word embedding semantic.
    - increase the value of the k parameter, the expected result will consist of more details about the final classification as each site will be represented by a higher number of variables.
- To compensate the full data-driven approach with final manual refinements

This experience showed that there is still a lot of work to be done before being able to produce meaningful results.

Annex 4: Proof-of-concept prototype **Experimental language statistics**

Overview (brief description)
A project to develop methodology to cluster businesses based on web scraped text data was undertaken by the Methodology division of the Office for National Statistics (ONS). Following ESSnet Big Data WP2 a dataset of web scraped data for registered companies in the UK was acquired. This data was used to cluster businesses with the aim of providing an alternative perspective to the Standard Industrial Classification. The methodology developed, which should be considered a work in progress, presupposes a large dataset of text referring to the activities of the businesses in question. For SIC comparison this data needs to also be linked to a register of businesses.
Data Sources (used data sources)
Four datasets were used in the project: acquired web scraped data, company information from Companies House, the Standard Industrial Classification of Economic Activities (SIC) for the UK, and pretrained word vectors. The web scraped data provided by a third-party consists of text extracted from the most likely candidate website from search engine queries for companies registered with Companies House. Each record has a score indicating a degree of confidence in the match between the business and the website. The text data is extracted from the HTML of the home page and an about page if it exists. This data was linked to company information from Companies House. Companies House provide publicly available data on companies registered in the UK. This formed the target population for the web scraped data. Companies House also provide SIC data which was used as a comparison for clustering produced by this project. SIC is identical to the Statistical Classification of Economic Activities in the European Community (NACE) to the four digit class level. The three components of the source data: web scraped text, company information and SIC are treated as a single dataset in the study. Pre-trained word vectors were used to improve cluster quality. Pre-trained word vectors from the Word2Vec model trained on part of the Google News dataset (see https://code.google.com/archive/p/word2vec/) are used to embed the web scraped text data into a word vector space of 300 dimensions.
Description (methods used, SW used, procedure adopted, decisions, etc....)
Software A command line application was created to run a pipeline to process and cluster the data with analysis and development done predominantly in Python. In addition to standard libraries for data analysis and visualisation (e.g. numpy, pandas, matplotlib) the following were used: • Natural Language Toolkit: https://www.nltk.org • gensim: https://radimrehurek.com/gensim/ • scikit-learn: https://scikit-learn.org • Keras: https://keras.io • Theano: http://deeplearning.net/software/theano/ • Uniform Manifold Approximation and Projection for Dimension Reduction: https://umap-learn.readthedocs.io

- The hdbscan Clustering Library (hdbscan): https://hdbscan.readthedocs.io

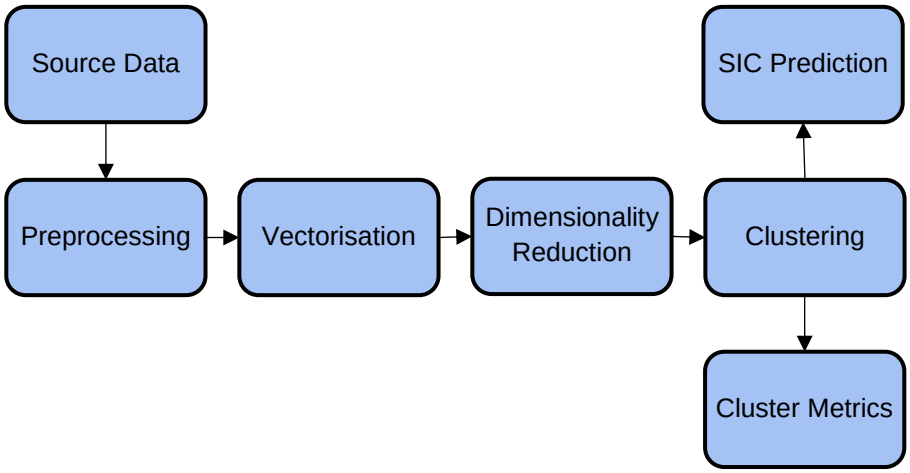
Procedure

Running the system requires an analyst to have a local copy of the data to cluster, along with a Python installation. The pipeline has been built and tested on a specific dataset, it is likely that some adjustment would be required to apply the same methods to a different data source. In particular the text cleaning steps and parameters for dimensionality reduction and clustering would need to be reviewed.

In Figure 1 the main stages of the processing pipeline are outlined. Data cleaning takes place in the preprocessing stage and includes lemmatisation of text. Dimensionality reduction is applied to the high dimensional vector embedding of the text data to improve clustering performance. The pipeline uses a density based clustering and relies on local structure being preserved by the dimensionality reduction.

Text data with associated website information is linked to businesses and has a score indicating certainty in the correctness of that link. To report metrics relative to SIC a threshold is applied to this score. Setting the threshold is a manual process which involves assessing a sample of links and creating a bootstrapped estimate of the ROC curve. This plot is used to choose a suitable threshold. It is worth noting that this procedure does not require that the score be well calibrated or even nominally a probability.

Figure 2: Outline of processing pipeline



Cluster validation and running a suite of metrics is largely a manual process. Validation consists of assessments of internal validity of clusters (i.e. based purely on cluster distribution in the lower dimensional embedding) as well as external validity (i.e. by comparison to SIC). The approaches can be further split into global and cluster local. Global level metrics such as homogeneity or differences in cluster size distribution give insight into overall comparability with SIC. Cluster local metrics such as impurity measures with respect to SIC allow areas which are a closer match to SIC to be identified. An exercise evaluating several models performing SIC prediction from clusters and the vector embeddings was also done.

Conclusions (lessons learned, open issues, obtained results, ...)

Conclusion

The methodology developed should be considered a work in progress. On manual inspection a significant proportion of the clusters do appear meaningful. However, the data used for clustering in the study was only a small fraction of the target population, with differences in the SIC class distribution. The clustering marks many businesses as outliers rather than associating them with a cluster. There has also not be enough work done to demonstrate the value the clustering adds to SIC to make a recommendation regarding adoption of the methodology.

A good ranking of uncertainty in linked source data allows decisions of trade-offs between type I and type II error to be made at the analysis stage of a pipeline. In this project having a score associated with the linkage of businesses to websites allowed analysis to take place despite data quality issues. In similar cases ranking quality should be assessed (for example by AUC or ROC plots) and maintained as new data is processed.

Topic analysis and clustering are an effective way of exploring large volumes of text data in a structured way. Initial data cleaning missed problems with data retrieval, for example when scrapers pulled text from error pages or cookie notifications. These formed distinct clusters that informed changes to the data cleaning process.

Desirable properties of the final clustering (e.g. stability, compactness, number of outliers or minimum cluster size) are influenced by the parameters of the pipeline. Production implementations should consider allowing configuration of these parameters for different use cases. During the project the ability to swap out algorithms in the pipeline was useful and it is recommended that any production system be designed with modularity in mind.

Lessons learned / open issues

Working with a fixed dataset rather than the results of a live system, while simplifying the scope of the project, did cause issues for the analysis. Data quality issues, such as error responses and problems reading JavaScript heavy websites, surfaced throughout the project. These could have been addressed if the data were coming from a live system. For example, by rerunning or improving the scraping code. As this project used a fixed dataset many records had to be dropped from the analysis.

Pretrained models are a powerful tool for language processing and further work may be needed to make best use of them within ONS. Multiple attempts were made to develop a vectorisation without relying on a pretrained model, however on manual inspection the clusters did not appear well formed. Using a reliable pretrained Word2Vec model with simple averaging of the resultant word vectors for cleaned website text improved subjective performance enough to make comparisons with SIC worthwhile.

It is desirable for each business in the population to be associated with a cluster. The main clustering approach adopted in this project produces a hard clustering, this simplifies cluster validation but is not ideal for economic activity which is a naturally fuzzy concept. The clustering also leaves many businesses unclustered. Demonstrating a clustering algorithm with fuzzy cluster membership and higher coverage with equivalent performance would be a good extension of the work done.

Further methodological work is required to identify the best metrics and post cluster validation checks to apply. Demonstrating relationships between clusters and variables linked to businesses would be a natural next step toward using the clustering for stratification.

Annex 5: Use Case 1 **URLs Inventory of enterprises**

Use Case Identification	
Use Case ID	CaseEnterpriseURLInventory
Use Case Name	URLs inventory of enterprises
Use Case Definition	
Description	The first task is to generate a URLs inventory for enterprises with ten or more employees (10+ employees) in the Statistical Business Register (SBR), and to update and maintain the URLs inventory. The Inventory will be used for web-crawling of the enterprise sites to retrieve information for variables in the ICT usage in enterprise survey and validation of SBR and NACE classification. The second task is to generate a URLs inventory for enterprises with less than ten employees (0-9 employees) in the Statistical Business Register, and to update and maintain the URLs inventory. The Inventory could be used for web-crawling of the enterprise sites to retrieve information for variables in the ICT usage in enterprise survey and validation of SBR and NACE classification. The two tasks are independent from each other and the execution of task 2 is optional. The approach for achieving the tasks is querying search engines for enterprises' URLs (e.g. Google, Bing etc.). The search queries are created from data from the SBR, such as enterprise name, address and/or other variables that improve the chance to find the right website. The search results contain a list of URLs and small text fragments, called snippets. The URLs (e.g. the top 10 links) and/or the yellow pages can be used by a web-scraper to scrape contents. Text analysis/matching to find the most relevant URL for the enterprise (e.g. looking for the enterprise's charatcteristics such as name, telephone number, addresses) is performed on the scraped pages and/or the snippets. The URLs Inventory of enterprises is created from the most relevant websites' URLs found. For updating and maintaining the URLs inventory and to check validity of URLs, this process needs to be repeated regularly. The frequency could be periodic e.g. once a year. Depending on the data update ambition the volume scale does not necessarily need to be the complete enterprise list, one could opt for a rotating sample. The search engine query program and web-scraper is an automated program to fetch information by URLs and to compare the retrieved information with the stored information in the SBR, such as address, name, identification number and etc. If the retrieved information is confirmed with the stored information, the URL is valid; otherwise, the URL is replaced by the URL with the highest confidence score or none if the score is below a threshold.

Actors	1. Program developer/Data Scientist: extract the urls from the available data sets; develop the programs and tools for URLs Retrieval; 2. URL inventory owner: maintain the inventory and keep the validity and the quality of the content in the inventory; 3. Statistician/Data Scientist: analyse the concept matching between web information and the Statistical Business Register and method problems such as missing values on the websites; 4. Statistical Business Register (SBR); 5. Big Data Ecosystem (e.g. system's software, databases, IT tools); 6. Search engines APIs; 7. Enterprise internet websites and Yellow pages websites; 8. Data (e.g. SBR, Search engines, Yellow pages, Enterprise internet websites).
Preconditions	1. ESS webscraping policy and/or websites' owners awareness; 2. Enterprise's characteristics for searching on Internet (name ID, phone, address, e-mail and etc.); 3. Available Search engines APIs and yellow pages websites (depending on volume and type of use one may need a paid API key); 4. Configured Big Data ecosystem; 5. A web scraper that scrapes the Internet according to the ethical rules; 6. Definition of suitable evaluation criteria and methods to assess enterprises' URLs and matching algorithms applying machine learning.
Postconditions	1. Links returned by search engine APIs and yellow pages; 2. Web content scraped by the links returned in step 1; 3. A URLs inventory with validated URLs of enterprises; 4. Quality indicators of the algorithms, URL inventory.
Frequency of Use	Every year in the period January - March.
Scenario	1. A software querying the search engine APIs and storing the returned features; 2. A web scraper downloads and stores the potential websites for analysing; 3. Feature extraction from the websites, building up the feature matrix for machine learning; 4. Machine learning to match the enterprises and URLs and/or deterministic decision; 5. Evaluating the result.
Special Requirements	Legal issues for web scraping on a large scale in Europe and data storage need to be regulated. A clear methodological framework is mandatory for using web scraping methods and web scraped data as a source for official statistics. In practice there is an ethical rule that is followed by many NSI's scrapers. The web scraper usually need not go deep than one level in the website. Usually this information is provided in contact section of the webpage or on main page.
Issues	Since it is regulated that only the relevant data should be stored for official statistics, the data generated in the process can be deleted when the URL repository is built. If the generic web scraping is not possible, extracting URLs from other available datasets can be an option. In such a case the study is focused on the method and classifier developing.

Annex 6: Use Case 2 **Variables in the ICT usage in enterprise survey**

Use Case Identification	
Use Case ID	2.1.
Use Case Name	Variables in the ICT usage in enterprise survey
Use Case Definition	
Description	The first goal of this use case is to predict whether an enterprise provides web sales facilities on its website. The general concept of this use case is that in order to achieve the goal a software has to scrape the webpages of an enterprises official website and then using machine learning techniques applied to texts (text mining) it should provide an accurate prediction. In the future more features could be added e.g. product catalogues or price lists, order tracking, advertisement of open job positions or online job applications. In some cases enterprises can have or can use more than one website. Therefore it is necessary to analyse all websites of the company and treat them as one website for analysis (making separate analysis of all the websites and provide several different results for one company is not recommended). The expected result of this use case is to make a decision whether the result can be used to substitute or replace questions from the ICT questionnaire or to use the new information to make even more detailed statistics on ICT use. The second goal of this use case is to provide information on the existence of a connection between the website and social media for a particular enterprise. It means that all social media accounts will be taken into account. We will especially search for Facebook and Twitter accounts as well as LinkedIn (Instagram; less likely). The general concept of this use case is to scrape the webpage and search for any links to Facebook/Twitter accounts. After that the Twitter or Facebook account can be verified for activity. In the future more attributes can be added e.g., whether the account is up to date and how often the content is changing. The risk is that some of the Facebook or Twitter links presented on webpages may be linked to other enterprises. This is the reason that evaluation of Facebook and Twitter profiles is necessary to provide reliable information. In some cases enterprises can have several Facebook profiles. Therefore it is necessary to link the main profile to the company. The expected results of this use case are to make a decision whether the data on social media activity can be used to enhance business registers. The third goal is to identify specific features of the website (e.g. product list, cookies). The fourth goal is to identify the purpose of the social media presence.

Actors	**Primary Actors**: 1. Statistician (Data Scientist). 2. Web crawler (web scraping tool). 3. NoSQL database. 4. (Potential) big data ecosystem (like Sandbox). 5. NSI data (Business Register, Administrative Data, Survey Data). 6. Enterprise internet website.
Preconditions	1. Validated list of URLs of enterprise websites. 2. Configured big data ecosystem. 3. Definition of suitable evaluation criteria or methods in order to assess if an enterprise does e-commerce, social media or specific feature of the website.
Postconditions	1. Success - positive or negative assessment (the enterprise does e-commerce or not, social media, specific feature of the website). 2. Failure - no access to the enterprise webpage.
Frequency of Use	Every year in the period of March and April
Scenario	1. Web crawler takes web address from Business Register. 2. Web crawler stores the enterprise internet webservice (webpage) in NoSQL database. 2. Algorithm looks for attributes of the webpage (social media link on the webpage, features of the website, e-commerce) 3. Algorithm identifies the feature - store the link in the NoSQL database and search the rest of the webpage. 4. Algorithm is finalising the web scraping. 5. If no links/features were identified in point 3, the result is also stored in the NoSQL database. 6. Statistician makes an analysis of reliability of information stored in NoSQL database.
Special Requirements	The webcrawler should not go outside the provided website domain and should avoid useless websites (e.g. portals that collect enterprise information like their phone numbers). The webcrawler should not go deeper than one level in the website. Usually this information is provided in contact section of the webpage or on main page (index.html/index.php/index.jsp etc.).
Issues	This use case needs access to a verified list of webpages.

Annex 7: Use Case 3 **Data driven discovery of emergent enterprise classifications**

Use Case Identification	
Use Case ID	3.1
Use Case Name	Data driven discovery of emergent enterprise classifications
Use Case Definition	
Description	Given a huge text corpus obtained by a massive web scraping activity on enterprise websites, natural language processing techniques (like, e.g. Word Embeddings, LDA, etc.) will be used to: 1) explore the semantic content of the corpus 2) identify meaningful semantic and syntactic patterns 3) use the discovered patterns to define data driven enterprise classifications 4) associate each enterprise with the most appropriate class of the discovered classifications The expected outcome is one or more new enterprise classifications and the corresponding distributions of the scraped enterprises.
Actors	**Primary Actors**: NSI researchers Web scraping platform Big data storage and analysis platform (e.g. Apache Solr, MongoDB, etc.) Word embedding algorithms (e.g. Word2Vec, GloVe, etc.) Topic modeling algorithms (e.g. LDA) Interactive graph-based exploration and visualization platform (Istat WordEmBox) Dimensionality reduction/clustering algorithms (e.g. PCA, t-SNE) **Supporting Actors**: NSI researchers in charge of mantaining the BR.
Preconditions	URL list of enterprises of interest
Postconditions	Describe the state of the system at the conclusion of the use case execution. Number each success and failure postcondition. 1) Success: the discovered classifications are meaningful and can be validated 2) Success: for each scraped enterprise a suitable class in the discovered classifications can be chosen 3) Failure: the discovered classifications are NOT meaningful and cannot be validated 4) Failure: the discovered classifications are too fuzzy so that it is impossible to uniquely associate an enterprise to a class
Frequency of Use	Every year during March - April.

Scenario	Main (Normal Course): Enter the main flow of events. This is best done as a numbered list of actions performed by the actor, alternating with responses provided by the system. Describe the flow of events from preconditions to postconditions when nothing goes wrong. 1) acquire an initial list of official enterprise URLs 2) massive webscraping of the textual content 3) storage of the scraped content in the big data storage platform 4) normalisation and cleaning 5) word embedding 6) word embedding and topic modeling for enterprise classification discovery
Special Requirements	No special requirements needed
Issues	The discovered classifications could be not meaningful and could not be validated The discovered classifications could be too fuzzy so that it will be impossible to uniquely associate an enterprise to a class

Annex 8: Use Case 4 **Natural Language Generated Enterprise Activity Clusters**

Use Case Identification	
Use Case ID	WPC.NaturalLanguageClusters
Use Case Name	Natural Language Generated Enterprise Activity Clusters
Use Case Definition	
Description	For the purpose of creating self-defined group memberships, e.g. automatic clustering of human language descriptions of enterprises to create groups by activity for use in stratifying industries. The outcome is information on membership of enterprises within a data-generated novel classification scheme. The goal in creating this output is to facilitate stratification of enterprises e.g. for survey purposes, and allow exploration of natural divisions in emerging business areas not adequately captured by traditional classification systems. Such a methodology may be used to cluster text from other pages to stratify enterprises by topics discussed, for example, topics found within pages on their websites related to "sustainability".
Actors	**Primary Actors**: NSI researcher/data scientist web crawler (gathers raw data/finds descriptions of companies online) NoSQL database (logical storage for document format) enterprise web pages (data source)
Preconditions	1. Method for linking websites to a list of enterprises with supporting information (primarily, NACE/SIC code). 2. Method for cleaning website text, or source of already-cleaned website text, on enterprise activities. 3. Sufficient quantity of data to support topic modelling methodology (requires >>10,000 records ideally). 4. Methods developed or researched to evalute cluster quality and coherency
Postconditions	Per enterprise: 1. Success - record of enterprise cluster membership linked to business register 2. Success - enterprise found not to have a website/identifiable web presence 3. Failure - enterprise website not accessible/available but exists Per study: 1. Success - meaningful clusters created from corpus of web text 2. Failure - clusters are meaningless or of low quality/poorly constrained membership
Frequency of Use	Once a year would be logical given the rate of change of the field of interest (industry types/concerns)
Scenario	Note: Steps 1 through 3 may be contracted/data purchased from 3rd party 1. Web crawler takes web address from NSI data (list of enterprise

	addresses) 2. Web crawler stores the Enterprise internet website content in NoSQL database. 3.a - The website content is parsed and cleaned to leave human-readable descriptions of company activities. 3.b - The website content is parsed and cleaned to leave human-readable descriptions of a sub-activity, e.g. actions on sustainability topics. 4. A topic model is generated using the human text, enterprises are assigned to these autonomously generated classifications. 5.a - Exception, poor quality clusters, these will be disaggregated and reassigned to other most probable clusters. 5.b - Enterprises that are edge cases or bad data are flagged for review by analyst.
Special Requirements	The webcrawler should not go outside the provided website domain and should avoid useless websites (e.g. portals that collect enterprise information like their phone numbers).
Issues	This use case needs access to verified list of webpages and to a list of domains to filter (typically national specific).

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